# TTRD3: Texture Transfer Residual Denoising Dual Diffusion Model for Remote Sensing Image Super-Resolution

Yide Liu, Haijiang Sun, Xiaowen Zhang, Qiaoyuan Liu, Zhouchang Chen, and Chongzhuo Xiao

***Abstract*—Remote sensing image super-resolution (RSISR) aims to reconstruct high-resolution (HR) remote sensing (RS) images from low-resolution (LR) RS image inputs by overcoming the physical limitations of imaging systems. As a specialized task in the RS domain, RSISR provides critical support for downstream fine-grained ground object interpretation. Traditional RSISR methods face three major technical shortcomings. First, since wide-area RS images exhibit highly complex ground object characteristics, it is challenging to extract effective multiscale features for SR reconstruction from their blurred representations. Second, the inherent imaging constraints of RS images lead to insufficient prior information, resulting in poor semantic consistency between SR reconstructions and real scenes. Third, existing SR methods struggle to balance the inherent tradeoff between geometric accuracy and visual quality. To address these challenges, this article proposes texture transfer residual denoising dual diffusion model (TTRD3), a diffusion model (DM)-based reference (Ref)-based super-resolution (RefSR) framework. Specifically, the multiscale feature aggregation block (MFAB) is designed to extract spatially heterogeneous features with scale variations through parallel heterogeneous convolutional kernels. The sparse texture transfer guidance (STTG) is developed to mine multiscale sparse texture features from high-resolution (HR) Ref images of similar scenes. The residual denoising dual diffusion model (RDDM) framework is constructed to synergistically optimize geometric accuracy and visual quality by integrating the deterministic modeling of residual diffusion with the diverse generation of noise diffusion. Experiments on multisource RS datasets demonstrate that TTRD3 significantly outperforms existing state-of-the-art (SOTA) methods in both geometric accuracy and visual quality. Our code and model are available at: https://github.com/LED-666/TTRD3**

***Index Terms*—Remote sensing, residual denoising dual diffusion model (RDDM), sparse texture transfer, super-resolution.**

## I. Introduction

REMOTE sensing technology has matured significantly, but its application remains constrained by the limitations of image spatial resolution in critical domains such as environmental monitoring [1], resource exploration [2], change detection [3], and national security [4]. SR technology compensates for the physical limitations of imaging systems through algorithmic enhancements, serving as a core solution to overcome spatial resolution bottlenecks. However, the intrinsic characteristics of RS images pose significant challenges. First, the complex interactions between ground objects (e.g., buildings and vegetation) make multiscale feature extraction difficult [5]. Second, irreversible high-frequency information loss during RS imaging and the scarcity of multisource paired reference (Ref) data lead to insufficient prior information [6]. Third, existing SR methods struggle to balance geometric fidelity and visual authenticity, often producing over-smoothed textures or repetitive unrealistic artifacts [7].

To address the challenge of feature extraction, current multiscale feature extraction methods often fail to capture the hierarchical features of RS images. PMSRN [8] employs dilated convolutions to expand receptive fields and uses a pyramidal multiscale residual network to enhance contextual interactions, effectively addressing multiscale ground object analysis. However, due to significant sampling intervals in convolutional kernels, it suffers from local information loss and limited performance in extracting small targets (e.g., roads and vehicles). WTCRR [9] utilizes wavelet transforms in the frequency domain to extract periodic textures and suppress noise, but lacks dynamic scale adaptation for multiscale targets in complex scenes. TransENet [10] improves long-range dependency modeling through Transformer [11] architectures but lacks hierarchical design in multiscale feature aggregation, leading to imbalanced feature weighting between small targets and large scenes.

Beyond enhancing feature extraction within a single image, another promising avenue to overcome the information loss is to leverage additional information from other sources. Ref-based super-resolution (RefSR) has emerged as a specialized paradigm that enhances the reconstruction quality by transferring high-frequency textures from auxiliary HR Ref images. Its inputs consist of a low-resolution (LR) image and semantically relevant Ref images (typically captured under different spatiotemporal conditions or viewpoints), while the

Received 18 May 2025; revised 21 September 2025; accepted 22 October 2025. Date of publication 27 October 2025; date of current version 12 November 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62475255, Grant 62535017, Grant 62401538, and Grant 62401539. *(Corresponding authors: Haijiang Sun; Qiaoyuan Liu.)*

Yide Liu, Xiaowen Zhang, Zhouchang Chen, and Chongzhuo Xiao are with Changchun Institute of Optics, Fine Mechanics and Physics, Chinese Academy of Sciences, Changchun 130033, China, and also with the University of Chinese Academy of Sciences, Beijing 100049, China (e-mail: liuyide23@mails.ucas.ac.cn; zhangxiaowen22@mails.ucas.ac.cn; chenzhouchang23@mails.ucas.ac.cn; xiaochongzhuo23@mails.ucas.ac.cn).

Haijiang Sun and Qiaoyuan Liu are with Changchun Institute of Optics, Fine Mechanics and Physics, Chinese Academy of Sciences, Changchun 130033, China (e-mail: sunhj@ciomp.ac.cn; liuqy@ciomp.ac.cn).

Digital Object Identifier 10.1109/TGRS.2025.3626248

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:58:10 UTC from IEEE Xplore. Restrictions apply.

output is a high-fidelity super-resolved image enriched with transferred textures. The core innovation lies in adaptive texture transfer: 1) calculating deep feature similarity between LR and Ref patches and 2) selectively transferring semantically aligned textures to compensate for irreversible high-frequency losses in LR inputs. Distinct from SISR, which relies solely on LR inputs and implicit priors from external datasets, RefSR explicitly leverages Ref images to provide texture-level guidance. Compared with unsupervised SR, RefSR utilizes external Ref supervision, achieving superior detail recovery when relevant Refs are available. Critically, while real-world LR images lack perfectly aligned HR counterparts, RefSR unlocks new potential by utilizing semantically equivalent Refs across spatiotemporal domains.

For instance, DI-GAN [12] and UPSNet [13] reconstruct HR spectral images using LR multispectral and panchromatic data. RGTGAN [14] leverages multiview RS images as Refs, while Ref-Diff [15] utilizes multitemporal RS image pairs and change detection masks as priors. However, these methods require strict spatial alignment between LR and Ref images, which are often unavailable in real-world scenarios. Although CycleGAN-based [16] unsupervised cross-modal fusion relaxes alignment constraints, it may introduce semantic inconsistencies. To address this, we explore sparse texture transfer from unpaired Ref images, injecting reasonably high-frequency components through similarity-guided diffusion without requiring pixel-level correspondence.

Existing SR frameworks face a dilemma: regression-based convolutional neural networks (CNNs) [17] and Transformer methods rely on deterministic approximations of LR-to-HR mappings, prioritizing geometric accuracy but producing blurred outputs [5]. Conversely, generative adversarial networks (GANs)-based [18] and diffusion model (DM) [19]-based methods enhance the visual quality at the cost of structural distortion. Specifically, GANs suffer from mode collapse, leading to repetitive textures [20], while DMs lack explicit geometric constraints, causing deformation of ground objects. Balancing geometric precision and visual quality remains an urgent challenge.

To address the aforementioned challenges, this study proposes the texture transfer residual denoising dual diffusion model (TTRD3), establishing a novel solution framework for RefSR tasks, as illustrated in Fig. 1. Specifically, to extract multiscale information from both Ref and LR images, we design a multiscale feature aggregation block (MFAB) employing parallel heterogeneous convolutional kernels, integrated with the convolutional block attention module (CBAM) to dynamically weight critical scale features, thereby significantly enhancing the feature representation capability for complex ground objects. To compensate for the inherent high-frequency information loss in remote sensing LR images, we additionally introduce Ref images from similar scenes. The sparse texture transfer guidance (STTG) is constructed to extract strongly correlated sparse texture features from Ref images through a soft-hard attention collaborative transfer mechanism. Here, soft attention models fine-grained texture similarity via local–global hybrid correlation matrices, while hard attention suppresses redundant noise through dynamic

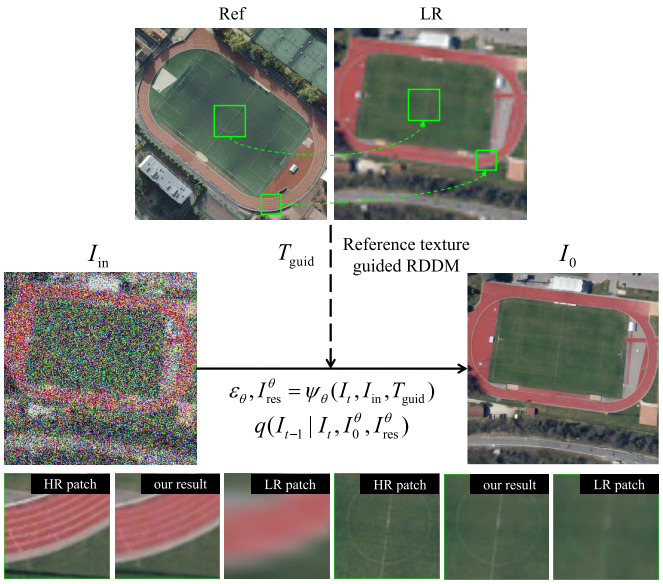



Fig. 1. TTRD3 framework extracts structurally similar high-frequency spatial texture information from Ref images, thereby guiding the RDDM to generate realistic high-frequency details. Here, Ref denotes the reference remote sensing image from a similar scene, while $T_{\text{guid}}$ represents the transferred texture guidance. The network $\psi_\theta$ predicts the noise $\varepsilon_\theta$ and residual information $I^\theta_{\text{res}}$, and reconstructs the SR image $I_0$ through the reverse process.

Top-$K$ sparse selection to improve the reconstruction accuracy in key regions. To resolve the challenging tradeoff between geometric precision and visual quality, we develop the residual denoising diffusion super-resolution architecture. Residual denoising dual diffusion model (RDDM) effectively balances the accuracy-authenticity dilemma in DM-based remote sensing SR tasks by combining the deterministic modeling of residual diffusion with the diverse generation capability of noise diffusion.

1) Design an MFAB feature extraction module to enhance the multiscale feature representation by integrating heterogeneous convolution and the CBAM attention mechanism.
2) Propose an STTG texture migration method that replaces the spatial alignment with deep semantic similarity, adaptively selecting strongly correlated sparse texture information to address the challenge of insufficient prior information in SR tasks.
3) Develop an RDDM-based SR framework that leverages deterministic residual diffusion to constrain the diversity of noise diffusion, achieving a balance between geometric accuracy and visual quality.

The remainder of this article is organized as follows. Section II provides an overview of the advancements in SR tasks. Section III elaborates on the technical details and highlights the innovations of our proposed TTRD3 framework. Section IV describes the datasets utilized and the experimental details. Section V presents the experiments and analysis. Finally, Section VI concludes this article with a summary of the key contributions and outlines potential future research directions.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:58:10 UTC from IEEE Xplore. Restrictions apply.

## II. Related Works

### A. Regression-Based SISR Models

With the advent of the deep learning revolution, regression-based deep SISR methods have gradually dominated the field. SRCNN [21] first constructed a three-layer convolutional network, breaking the performance bottleneck of traditional algorithms, but its shallow structure struggled to capture deep-level features. VDSR [22] overcame network depth limitations through residual learning, enabling deeper SR networks. The introduction of the ResNet [23] architecture gave rise to deep residual networks like SRResNet [24], achieving higher performance metrics. SRDenseNet [25] adopted dense connections to enhance feature reuse. DRCN [26] and DRRN [27] employed recursive structures to increase network depth while reducing computational complexity and storage costs through parameter sharing. CARN [28] built a lightweight recursive model for more efficient modeling. Although CNN-based methods have been widely implemented and achieved outstanding performance across tasks, their equal treatment of channels via convolutional kernels limits the utilization of high-frequency information. Attention mechanisms emerged as a solution, enabling deeper interdependencies among channels. RCAN [29] introduced channel attention modules (CAMs) that assign varying weights to each channel by learning global information from input feature maps. SAN [30] proposed a channel attention mechanism based on second-order statistics, enhancing feature correlations for more discriminative representations. The introduction of Transformer architectures brought breakthroughs in global modeling: SwinIR [31] achieved efficient long-range dependency capture through hierarchical window self-attention. TTSR [32] pioneered Ref image-guided paradigms to mitigate high-frequency information loss. ESRT [33] balanced performance and computational costs via a CNN–Transformer hybrid architecture, and ELAN [34] optimized attention-sharing mechanisms to improve parameter efficiency. In remote sensing image super-resolution (RSISR) tasks, ERCNN [35] proposed a dual-brightness scheme to optimize feature retention and cross-feature map discriminability. FGRDN [36] employed novel feedback mechanisms and ghost modules for low-parameter feature refinement.

Recent studies have made further progress in addressing computational efficiency and multiscale representation within regression frameworks. ESTNet [37] proposes a method reducing parameters by 82.68% via channel attention pruning, significantly enhancing deployment capability on edge devices. Meanwhile, MFG-HMoE [38] employs a heterogeneous mixture-of-experts architecture with a dual-routing mechanism that adaptively activates specialized upsampling paths per pixel, enabling precise modeling of diverse ground objects.

MAST [39] designed multiscale attention to enhance geographical feature perception, while TTST [40] established a token selection mechanism to dynamically choose Top-$k$ critical tokens for more effective and compact long-range modeling. However, these regression methods centered on $L1$ (mean absolute error) and $L2$ (mean square error) loss functions, while optimizing image reconstruction accuracy through pixel-level difference minimization, inherently force generated results to converge toward "average solutions" of target images, leading to high-frequency texture loss and edge blurring—particularly evident in complex scenes like foliage and architectural textures [41].

### B. GAN-Based SISR Models

In recent years, GANs have achieved remarkable progress in SISR, opening new technical pathways to break through the limitations of traditional regression methods. SRGAN [24] first introduced adversarial training to SR, synthesizing highly realistic textures through generator-discriminator game mechanisms, significantly enhancing visual quality. ESRGAN [42] improved upon SRGAN by designing residual-in-residual dense blocks (RRDBs) and removing batch normalization layers, while introducing perceptual loss functions that leverage pretrained VGG models to align generated images with real ones in feature space. Real-ESRGAN [43] extended ESRGAN to handle real-world degradation patterns. SAGAN [44] integrated cross-image attention mechanisms to establish long-range, multilevel dependencies for detail generation. LDL [45] enhanced image authenticity through local discriminative learning to distinguish GAN-generated artifacts from real details. For RSISR tasks, SCSE-GAN [46] stacked spatial and channel squeeze–excitation (SCSE) blocks after residual layers and adopted Wasserstein GAN with gradient penalty (WGAN-GP) to stabilize training. MA-GAN [47] developed a multiattention framework with attention-based fusion blocks for improved geographical feature representation. SRAGAN [48] optimized adversarial learning by capturing structural components and long-term dependencies through local and global attention mechanisms. TE-SAGAN [49] introduced self-attention modules and texture loss units to refine edges and textures. Although GAN-based methods excel in perceptual metrics like learned perceptual image patch similarity (LPIPS), and hybrid loss strategies have substantially improved texture generation, challenges remain in computational complexity and training stability. DMs, with their progressive generation paradigm, offer new solutions by achieving high-quality reconstruction without mode collapse, emerging as a promising direction for next-generation SR technology.

### C. Diffusion-Based SISR Models

DMs establish a rigorous mathematical framework for image generation through dual processes: forward noise perturbation and reverse denoising learning. DDPM [50], as a milestone work, first proposed a trainable Markov chain diffusion architecture. DDIM [51] introduced non-Markovian accelerated sampling strategies to significantly improve inference efficiency. RDDM [52] changes the traditional diffusion paradigm by additionally introducing residual information to constrain the generation results, achieving remarkable effects in the fields of generation, image dehazing, and denoising. Classifier guidance [53] and classifier-free guidance [54] techniques achieved comprehensive quality superiority over GANs through conditional control mechanisms. Stable diffusion [55] addresses the computational costs and accuracy constraints

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:58:10 UTC from IEEE Xplore. Restrictions apply.

of previous approaches by developing a latent DM (LDM), which mitigates the challenges of building DMs directly on high-dimensional feature spaces. Compared with traditional GANs and VAEs, DMs demonstrate significant advantages in modeling complex image statistical distributions, with their progressive generation characteristics effectively avoiding mode collapse.

SR3 [56] pioneered the application of DMs to SR tasks, achieving HR image reconstruction through end-to-end noise prediction. SRDiff [57] proposed a residual diffusion paradigm by predicting LR–HR residual mappings, and ResShift [58] subsequently built an efficient residual chain generation architecture. For the RS images' characteristics, TESR [59] developed a two-stage enhancement SR framework, first validating the feasibility of DMs in RSISR. EDiffSR [60] innovatively designed a conditional prior enhancement module (CPEM) and efficient activation network (EANet), substantially reducing computational complexity. FastDiffSR [61] constructed a conditional diffusion acceleration framework, achieving breakthrough improvements in inference speed.

Beyond standard DMs, LDMs have emerged as a computationally efficient paradigm that shifts the diffusion process from pixel space to a compressed latent space. This approach significantly reduces computational costs while preserving high-frequency details, making it particularly suitable for HR remote sensing imagery. StableSR [62] pioneers the exploitation of a frozen stable diffusion [55] prior for real-world image super-resolution by inserting a lightweight time-aware encoder that modulates the generative denoising process at every step, achieving enhanced fidelity and detail without retraining the large-scale diffusion backbone. In RSISR tasks, LDCSR [63] proposes compressing the diffusion steps to just 4 steps by encoding differential priors between LR and LR, substantially improving inference speed. LDSR-S2 [64] leverages encoded LR images to guide the diffusion process, enhancing spectral consistency in multispectral Sentinel-2 imagery and providing pixel-level uncertainty quantification for the first time. Furthermore, RSDiffSR [65] extends the application of LDMs to RSISR. It mitigates the distribution gap between natural and remote sensing images via low-rank adaptation (LoRA) techniques and designs an enhanced control mechanism to constrain the geometric fidelity of generated details.

However, most existing DM-based remote sensing image SR tasks focus on designing novel conditional guidance modules and improving noise predictors. While these approaches optimize network architectures and achieve higher quantitative metrics, they fail to fundamentally enhance the noise prediction paradigm of DMs. This oversight allows the stochastic nature of noise diffusion to introduce semantic inconsistencies between generated results and ground truth. Similarly, severe information loss in LR images makes it challenging to extract sufficient high-frequency details solely through enhanced conditional guidance, further limiting the model's ability to recover authentic textures. These issues collectively degrade geometric accuracy, constraining the practical application of DM-based SR models.

### D. Ref-Based Sr Models

Unlike SISR that solely relies on LR inputs, RefSR significantly enhances reconstruction quality by leveraging similar texture information from Ref images to acquire additional priors. Although real-world LR images often lack perfectly aligned HR counterparts, equivalent HR images captured under different spatiotemporal conditions or camera viewpoints can still provide semantically relevant texture information, offering critical prior support for SR. These HR images, termed Ref images, enhance SR reconstruction through texture transfer mechanisms, forming the core workflow of RefSR. This process transfers supplementary texture details to the super-resolved LR image, a methodology defined as RefSR.

SRNTT [66] pioneered the first GAN-based RefSR approach, introducing a Ref-guided patch matching strategy that utilizes local texture patches and establishes multilevel patch matching to fully exploit Ref image information. Its key contribution lies in computing texture similarity not based on raw pixels but on multilevel image features extracted from a pretrained VGG network, which are subsequently fused. Inspired by SRNTT, TTSR [32] proposed a Transformer-like architecture that discovers deep feature correspondences through attention mechanisms, enabling accurate texture transfer for superior results. MASA-SR [67] addressed color and luminance discrepancies between LR and Ref images by learning distribution differences and applying spatially adaptive adjustments, significantly enhancing network robustness. In a different vein, CoSeR [68] circumvents the dependency on external Ref images by synthesizing high-fidelity Ref content directly from the LR input. It integrates language-guided diffusion priors with image appearance cues, and further introduces an "all-in-attention" conditional injection mechanism to improve the fidelity and semantic consistency of the generated Refs.

In the RS domain, RefSR methods typically utilize multisource Refs, such as multiview, multitemporal, or cross-modal data (e.g., infrared and panchromatic bands). RRSGAN [69], one of the earliest RefSR networks in RS, introduced a correlation attention module to strengthen feature alignment robustness. Recently, RGTGAN [14] proposed a gradient-assisted texture enhancement GAN, designing a gradient-guided module to improve high-frequency structure learning and employing deformable convolutions to align heterogeneous LR-Ref features. SpecGAN [70] represents a notable advancement by uniquely exploiting hyperspectral images (HSIs) as conditional inputs. Its dual-branch network effectively fuses LR-RGB and HSI features, leveraging the spectral specificity of materials to ensure exceptional semantic accuracy in large-factor ($32\times$) SR. SGDM [71] utilizes a pretrained stable DM as an implicit prior. To reduce hallucinations, it integrates vector maps for explicit structural guidance and models sensor-specific imaging characteristics through normalizing flows. This approach enables style-adaptive SR without requiring pixel-aligned Refs.

Despite advancements in texture transfer, existing methods predominantly rely on GAN frameworks, which are prone to local optima, often generating repetitive structures or semantic distortions that degrade downstream task performance.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:58:10 UTC from IEEE Xplore. Restrictions apply.

To overcome this bottleneck, our approach dynamically balances geometric accuracy and visual quality by constraining stochastic noise diffusion with residual-based deterministic modeling. Concurrently, the STTG mechanism adaptively selects semantically aligned high-frequency textures from HR Refs, providing complementary prior guidance to the RDDM framework, jointly ensuring semantic consistency and detail authenticity in reconstructions.

## III. Methods

### A. Residual Denoising Dual DM

To address the special requirements of reconstruction fidelity in RSISR tasks, the RDDM simultaneously diffuses residual information and noise information during the forward diffusion process. Specifically, residual diffusion emphasizes determinism by constraining the spatial fidelity of the reconstruction process through directional propagation of residual information. The noise diffusion focuses on diversity, representing stochastic perturbations in the diffusion process to ensure reconstruction diversity. The forward diffusion process and reverse inference process of RDDM will be elaborated in detail as follows.

*1) Forward Diffusion Process:* The forward process of RDDM allows for the simultaneous diffusion of residuals and noise, with the target image gradually diffusing into a degraded version carrying noise, where a single forward diffusion step is defined as follows:

$$I_t = I_{t-1} + I^t_{\text{deg}}, \quad I^t_{\text{deg}} \sim \mathcal{N}\left(\alpha_t I_{\text{res}}, \beta_t^2 \mathrm{I}\right). \tag{1}$$

Here, $I^t_{\text{deg}}$ represents the degradation information from state $I_{t-1}$ to state $I_t$, which includes residual diffusion and noise diffusion. $I_{\text{res}}$ is the difference between the input image $I_{\text{in}}$ and the original image $I_0$

$$I_{\text{res}} = I_{\text{in}} - I_0. \tag{2}$$

The forward process is somewhat similar to the residual diffusion in ResShift [58]. The key difference, however, is our decoupling of the diffusion processes. RDDM employs independent schedules, $\alpha_t$ and $\beta_t^2$, for residual and noise, moving beyond ResShift's coupled approach. The specific method is as follows:

$$\begin{aligned} I_t &= I_{t-1} + \alpha_t I_{\text{res}} + \beta_t \varepsilon_{t-1} \\ &= I_{t-2} + (\alpha_{t-1} + \alpha_t) I_{\text{res}} + \left(\sqrt{\beta_{t-1}^2 + \beta_t^2}\right) \varepsilon_{t-2} \\ &= \cdots \\ &= I_0 + \bar{\alpha}_t I_{\text{res}} + \bar{\beta}_t \varepsilon. \end{aligned} \tag{3}$$

Here, $\varepsilon_t, \varepsilon_{t-1}, \ldots, \varepsilon \sim \mathcal{N}(0, \mathrm{I})$; $\bar{\alpha}_t = \sum_{i=1}^t \alpha_t$; $\bar{\beta}_t = \left(\sum_{i=1}^t \beta_i^2\right)^{1/2}$. If $t = T$, then $\bar{\alpha}_T = 1$ and $I_T = I_{\text{in}} + \bar{\beta}_T \varepsilon$. Among them, $\bar{\beta}_T$ is a manually set parameter used to control the intensity of noise.

From (1), it can be seen that the joint probability distribution of the forward diffusion process is defined as follows:

$$q\left(I_{1:T} | I_0, I_{\text{res}}\right) = \prod_{t=1}^{T} q\left(I_t | I_{t-1}, I_{\text{res}}\right) \tag{4}$$

$$q\left(I_t | I_{t-1}, I_{\text{res}}\right) = \mathcal{N}\left(I_t; I_{t-1} + \alpha_t I_{\text{res}}, \beta_t^2 \mathrm{I}\right). \tag{5}$$

Equation (3) defines the marginal probability distribution as follows:

$$q\left(I_t | I_0, I_{\text{res}}\right) = \mathcal{N}\left(I_t; I_0 + \alpha_t I_{\text{res}}, \beta_t^2 \mathrm{I}\right). \tag{6}$$

*2) Reverse Inference Process:* In the forward process, the residual ($I_{\text{res}}$) and noise ($\varepsilon$) are progressively added to $I_0$ through successive diffusion steps to form $I_t$, with this cumulative transformation represented by the direct mapping from $I_0$ to $I_t$ in equation (3). During the reverse reconstruction process from $I_T$ to $I_0$, the network estimates $I_{\text{res}}$ and $\varepsilon$ for subtraction. $I_{\text{res}}$ and $\varepsilon$ can be estimated by the network. Training networks and using the input images $I_t$ and $I_t$ can be synthesized using $I_0$, $I_{\text{res}}$, and $\varepsilon$ by (3).

The target image $I_0^\theta$ can be estimated according to (3)

$$I_0^\theta = I_t - \bar{\alpha}_t I_{\text{res}}^\theta - \bar{\beta}_t \varepsilon^\theta. \tag{7}$$

Here, $I_0^\theta$ and $I_{\text{res}}^\theta$ can be predicted by the neural network, the reverse inference process is defined as follows:

$$p_\theta\left(I_{t-1} | I_t\right) = q_\sigma\left(I_{t-1} | I_t, I_0^\theta, I_{\text{res}}^\theta\right) \tag{8}$$

where the transition probability from $I_t$ to $I_{t-1}$ is

$$\begin{aligned} q_\sigma\left(I_{t-1} | I_t, I_0, I_{\text{res}}\right) = \mathcal{N}\Big(I_{t-1}; I_0 + \bar{\alpha}_{t-1} I_{\text{res}} \\ + \sqrt{\bar{\beta}_{t-1}^2 - \sigma_t^2} \frac{I_t - (I_0 + \bar{\alpha}_{t-1} I_{\text{res}})}{\bar{\beta}_t}, \sigma_t^2 \mathrm{I}\Big) \end{aligned} \tag{9}$$

where $\sigma_t^2 = \eta \beta_t^2 \bar{\beta}_{t-1}^2 / \bar{\beta}_t^2$, and $\eta$ controls the randomness ($\eta = 1$) or determinacy ($\eta = 0$) of the generation process.

From (3) and (8), it can be seen that (9) can be written as follows:

$$\begin{aligned} I_{t-1} &= I_0 + \bar{\alpha}_{t-1} I_{\text{res}}^\theta \\ &\quad + \sqrt{\bar{\beta}_{t-1}^2 - \sigma_t^2} \frac{I_t - \left(I_0 + \bar{\alpha}_t I_{\text{res}}^\theta\right)}{\bar{\beta}_t} + \sigma_t \varepsilon_t \\ &= \frac{\sqrt{\bar{\beta}_{t-1}^2 - \sigma_t^2}}{\bar{\beta}_t} I_t + \left(1 - \frac{\sqrt{\bar{\beta}_{t-1}^2 - \sigma_t^2}}{\bar{\beta}_t}\right) I_0 \\ &\quad + \left(\bar{\alpha}_{t-1} - \frac{\sqrt{\bar{\alpha}_t \bar{\beta}_{t-1}^2 - \sigma_t^2}}{\bar{\beta}_t}\right) I_{\text{res}}^\theta + \sigma_t \varepsilon_t \\ &= I_t - (\bar{\alpha}_t - \bar{\alpha}_{t-1}) I_{\text{res}}^\theta - \left(\bar{\beta}_t - \sqrt{\bar{\beta}_{t-1}^2 - \sigma_t^2}\right) \varepsilon_\theta + \sigma_t \varepsilon_t \end{aligned} \tag{10}$$

where $\varepsilon_t \sim \mathcal{N}(0, I)$. Equation (10) is the sampling formula for the reverse process.

Similar to the evolution from DDPM to DDIM, an inductive argument from time $T$ to time 1 can be used here to prove that the following equation holds at any time $t$

$$q\left(I_t | I_0, I_{\text{res}}\right) = \mathcal{N}\left(I_t; I_0 + \bar{\alpha}_t I_{\text{res}}, \bar{\beta}_t^2 \mathrm{I}\right). \tag{11}$$

Therefore, the final reverse sampling formula is:

$$I_{\text{prev}} = I_t - \left(\bar{\alpha}_t - \bar{\alpha}_{\text{prev}}\right) I_{\text{res}}^\theta - \left(\bar{\beta}_t - \sqrt{\bar{\beta}_{\text{prev}}^2 - \sigma_t^2}\right) \varepsilon_\theta + \sigma_t \varepsilon. \tag{12}$$

Among them, there can be multiple iteration steps between $I_{\text{prev}}$ and $I_t$.

A schematic of the forward process and reverse inference of the RDDM is shown in Fig. 2.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:58:10 UTC from IEEE Xplore. Restrictions apply.

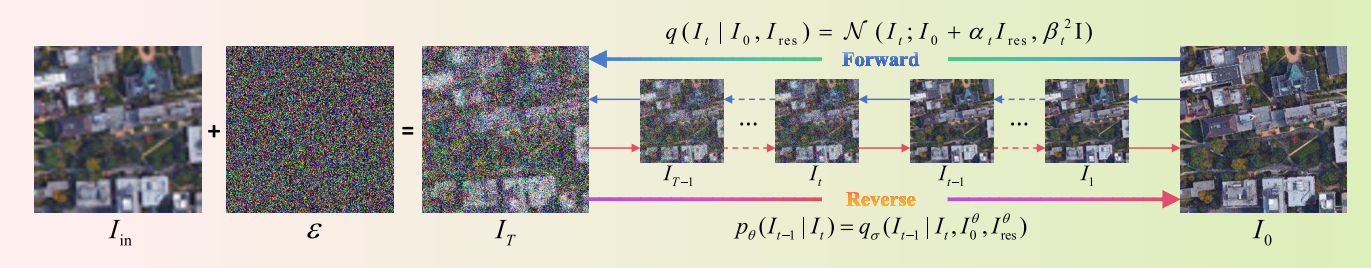



Fig. 2. Forward process and reverse inference process of the RDDM, comprising two components: residual diffusion and noise diffusion. Here, $I_0$ represents the HR image, $I_T$ denotes the dual-diffusion degraded image, $I_{in}$ corresponds to the residual degraded image (LR image), and $\varepsilon$ signifies the Gaussian noise.

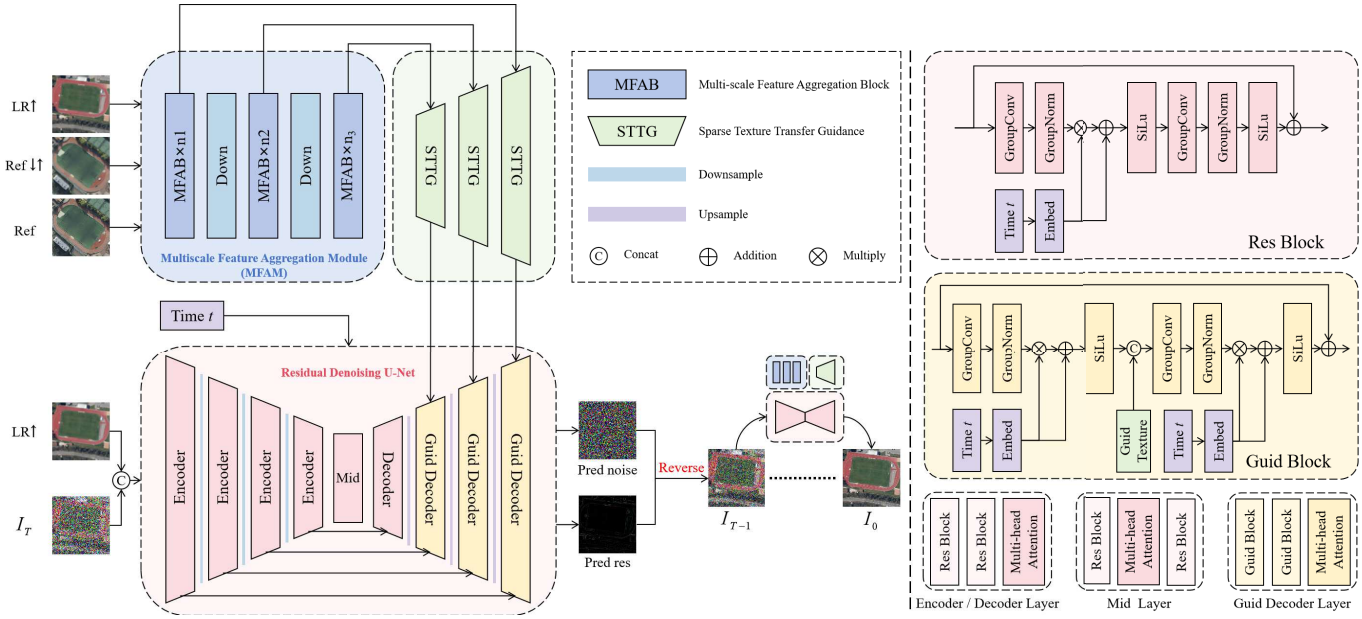



Fig. 3. Overall framework of the proposed TTRD3 is illustrated. Here, LR↑ denotes the 4× upsampled LR image, Ref↓↑ represents the degraded Ref image obtained by downsampling and subsequent upsampling, and Ref is the high-quality Ref image. These inputs are processed by the MFAM to extract multiscale features at three levels. The STTG then generates multiscale sparse texture guidance maps from these features. Finally, the guided decoder integrates the texture guidance information into the residual denoising U-Net network for high-fidelity reconstruction.

### B. Overview

The workflow of the TTRD3 network framework is illustrated in Fig. 3. The network inputs include the bicubic-interpolated LR remote sensing image LR↑, the original Ref image Ref, and its degraded version Ref↓↑. These inputs are first processed by the multiscale feature aggregation module (MFAM), which employs a pyramid architecture to hierarchically extract multiscale ground object features (e.g., buildings and vegetation) through the parallel multiscale convolutional kernels in the MFAB. This hierarchical extraction ultimately generates three distinct scales of deep feature maps.

Subsequently, the upsampled LR↑ image and the multiscale features from the Ref image group are fed into the STTG module. This module constructs local–global similarity matrices between LR and Ref feature patches using a soft-hard attention collaborative mechanism, combined with a dynamic Top-$K$ selection strategy to filter strongly correlated semantic textures. This process generates multiscale sparse texture guidance maps.

In the RDDM stage, the texture guidance information is injected into the residual denoising U-Net network via the Guid Decoder layer. During the diffusion process, the LR image and the residual noise-degraded image $I_T$ are used as conditional inputs, jointly with the texture guidance information, to predict the residual and noise components. Finally, the high-fidelity SR image is reconstructed through an iterative reverse diffusion process that removes noise and fuses residual information.

Notably, to better balance performance and computational efficiency, we designed three network variants with varying capacities (TTRD3-1Net-A/B and TTRD3-2Net). Detailed performance comparisons and efficiency analyses will be comprehensively discussed in Section V.

### C. Multiscale Feature Aggregation Block

To address multiscale inconsistency between Ref and LR images, we design the MFAB whose architecture is detailed in Fig. 4. This block employs parallel heterogeneous convolutional kernels to extract hierarchical spatial features, and integrates the CBAM—comprising a CAM and spatial attention module (SAM) as annotated in Fig. 5 to dynamically recalibrate cross-scale feature representations. This

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:58:10 UTC from IEEE Xplore. Restrictions apply.

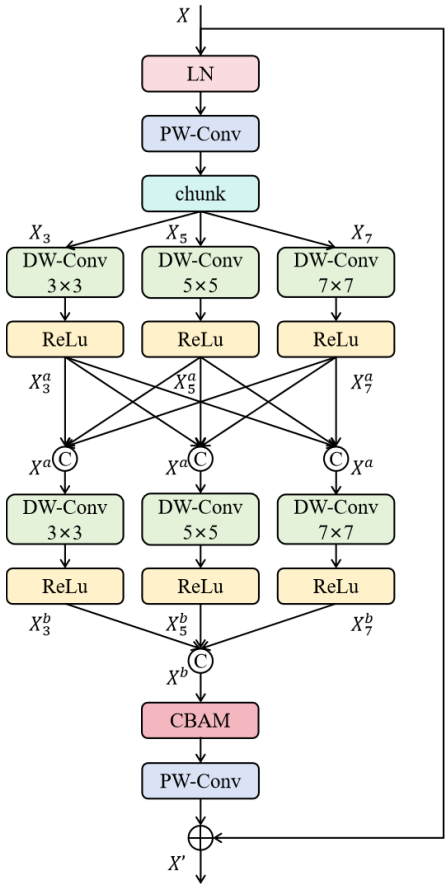



Fig. 4. Illustrates MFAB.

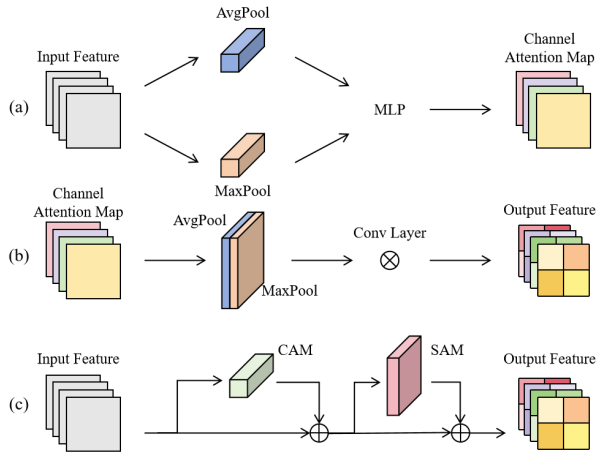



Fig. 5. Structural diagram of CBAM. (a) CAM, (b) SAM, and (c) overall process of CBAM.

dual-attention mechanism effectively mitigates feature fusion conflicts while suppressing high-frequency noise in complex remote sensing scenes.

Specifically, given an input feature tensor $X \in \mathbb{R}^{C\times H\times W}$, the process begins with layer normalization (LN) followed by a pointwise (PW) convolution to adjust the channel dimension to $3 \times C'$. The tensor is then split along the channel dimension into three subtensors ($X_3$, $X_5$, $X_7$), each with $C'$ channels

$$X_3, X_5, X_7 = \text{chunk}\left(f_{1\times1}\left(\text{LN}\left(X\right)\right)\right). \tag{13}$$

Each subtensor is processed by a $3 \times 3$, $5 \times 5$, or $7 \times 7$ depthwise (DW) convolution to extract multiscale local features. The processed features are then concatenated to generate the shallow fused features $X^a$.

$$\begin{cases} X_3^a = \text{ReLU}\left(f_{3\times3}\left(X_3\right)\right) \\ X_5^a = \text{ReLU}\left(f_{5\times5}\left(X_5\right)\right) \\ X_7^a = \text{ReLU}\left(f_{7\times7}\left(X_7\right)\right) \end{cases}$$
$$X^a = \text{Concat}\left(X_3^a, X_5^a, X_7^a\right). \tag{14}$$

Further enhanced through secondary multiscale DW convolutions, the deep multiscale features ($X_3^b$, $X_5^b$, $X_7^b$) are aggregated into $X^b$. $X^b$ is adaptively weighted by CBAM, which adjusts spatial and channelwise importance across different scales. Finally, a PW convolution restores the original channel dimension $C$, and a residual connection preserves the input's original information to generate the output $X'$

$$\begin{cases} X_3^b = \text{ReLU}\left(f_{3\times3}\left(X^a\right)\right) \\ X_5^b = \text{ReLU}\left(f_{5\times5}\left(X^a\right)\right) \\ X_7^b = \text{ReLU}\left(f_{7\times7}\left(X^a\right)\right) \end{cases}$$
$$X^b = \text{Concat}\left(X_3^b, X_5^b, X_7^b\right)$$
$$X' = X + f_{1\times1}\left(f_{\text{CBAM}}\left(X^b\right)\right). \tag{15}$$

### D. Sparse Texture Transfer Guidance

This study proposes an STTG method, its core innovation lies in constructing a collaborative framework combining a local–global hybrid soft attention mechanism and a learnable Top-$K$ sparsification hard attention mechanism, as shown in Fig. 6, enabling end-to-end cross-modal texture matching and transfer.

*1) Multiscale Semantic Embedding:* To eliminate scale discrepancies between the Ref image and the LR image LR, Ref is processed via bicubic downsampling-upsampling to obtain Ref $\downarrow\uparrow$, while LR is upsampled to generate LR $\uparrow$. The preprocessed images and the original HR Ref are fed into the MFAM in parallel to obtain three multiscale deep features $f^s_{\text{MFAM}}(\text{LR}\uparrow)$, $f^s_{\text{MFAM}}(\text{LR}\uparrow)$, and $f^s_{\text{MFAM}}(\text{LR}\uparrow)$, here $s \in \{1, 2, 3\}$. The preprocessed dual-modal data and the original HR Ref are fed into the MFAM. These feature maps are processed through patch unfolding operations to extract local texture representations. The resulting query, key, and value sequences are

$$\begin{cases} Q^s = f_{\text{unfold}}\left(f^s_{\text{MFAM}}\left(\text{LR}\uparrow\right)\right) \\ K^s = f_{\text{unfold}}\left(f^s_{\text{MFAM}}\left(\text{Ref}\downarrow\uparrow\right)\right) \\ V^s = f_{\text{unfold}}\left(f^s_{\text{MFAM}}\left(\text{Ref}\right)\right). \end{cases} \tag{16}$$

Each resulting tensor has the shape $\mathbb{R}^{B\times(C^s\cdot k^2)\times P^s}$, where $P^s = H^s \times W^s$ is the number of patches, $C^s$ is the channel dimension which varies across scales, and $k$ is the kernel size of the unfolding operation.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:58:10 UTC from IEEE Xplore. Restrictions apply.

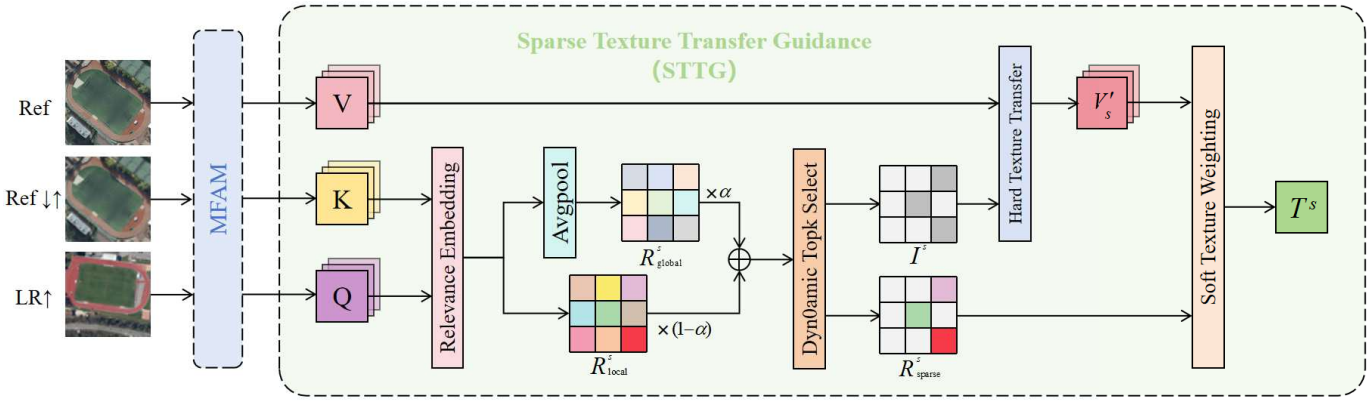



Fig. 6. STTG framework captures local details of texture while maintaining global perceptual consistency through a hybrid local–global soft attention mechanism. It dynamically balances the weights of global attention using a learnable parameter $\alpha$ and retains sparse correlation matrices and corresponding indices through a learnable Top-$K$ hard attention mechanism. By combining soft and hard attention via matrix multiplication, it achieves adaptive texture transfer.

*2) Local–Global Correlation Modeling:* The local attention mechanism captures fine-grained texture correspondences between the Ref image and the upsampled LR image (LR ↑). We compute the local correlation matrix $R^s_{\text{local}}$ from the normalized query and key features

$$\hat{Q}^s = \frac{Q^s}{\|Q^s\|_2}, \quad \hat{K}^s = \frac{K^s}{\|K^s\|_2}$$
$$R^s_{\text{local}} = \hat{Q}^s \cdot \left(\hat{K}^s\right)^\top. \tag{17}$$

Complementarily, the global attention mechanism models scene-level semantic compatibility. This is achieved by first condensing the feature sequences into global vectors via average pooling, and then, broadcasting their similarity score to form a global correlation matrix $R^s_{\text{global}}$

$$\bar{Q}^s = \frac{1}{N^s}\sum_{i=1}^{N^s} \hat{Q}^s_i, \quad \bar{K}^s = \frac{1}{N^s}\sum_{j=1}^{N^s} \hat{K}^s_j$$
$$R^s_{\text{global}} = \left(\bar{Q}^s \cdot \left(\bar{K}^s\right)^\top\right) \cdot \mathbf{1}_{N^s \times N^s} \tag{18}$$

where $\mathbf{1}_{N^s \times N^s}$ is an all-one matrix.

The final hybrid attention matrix $R^s_{\text{mix}}$ is a weighted fusion of local and global correlations via a learnable parameter $\alpha$

$$R^s_{\text{mix}} = \alpha \cdot R^s_{\text{global}} + (1-\alpha) \cdot R^s_{\text{local}}. \tag{19}$$

*3) Dynamic Top-K Sparsification:* To adaptively sparsify the dense correlation matrix $R^s_{\text{mix}}$, we introduce a learned Top-$K$ selection mechanism using the Gumbel-Softmax trick and a Straight-Through Estimator (STE) [72]. This process outputs a sparse matrix $R^s_{\text{sparse}}$ that retains only the most relevant connections.

*a) Dynamic K-Value Selection:* A learnable parameter vector $W^s_k$ predicts a distribution over possible sparsity levels. From this, we sample a discrete value $K^s \in [1, K_{\max}]$ in a differentiable manner

$$P^s = \text{Gumbel-Softmax}\left(W^s_k, \tau\right)$$
$$k_{\text{hard}} = \arg\max\left(P^s\right) + 1$$
$$k_{\text{soft}} = \sum_{k=1}^{K_{\max}} P^s_k \cdot k$$
$$K^s = k^{\text{det}}_{\text{hard}} + k_{\text{soft}} - k^{\text{det}}_{\text{soft}} \tag{20}$$

where $W^s_k$ is a learnable parameter vector, $\tau$ denotes temperature, $P^s$ is the probability distribution output by Gumbel-Softmax, and $K_{\max}$ is the maximum sparsity level. The superscript "det" denotes the gradient detachment operation, ensuring an integer $K^s$ is used in the forward pass, while gradients flowthrough the continuous approximation $k_{\text{soft}}$ during backpropagation.

*b) Sparse Matrix Construction:* For each query position $i$, we preserve only the Top-$K^s$ values from its row in $R^s_{\text{mix}}$, setting all others to 0

$$R^s_{\text{sparse}}(i,j) = \begin{cases} R^s_{\text{mix}}(i,j), & \text{if } j \in \text{TopK}\left(R^s_{\text{mix}}(i,:), K^s\right) \\ 0, & \text{otherwise.} \end{cases} \tag{21}$$

Here, $\text{TopK}(\cdot, K^s)$ returns the indices of the $K^s$ largest elements. This yields the final sparse guidance map.

*4) Soft-Hard Attention Collaborative Transfer:* Target-oriented texture features are extracted through pointwise multiplication between the sparse correlation matrix $R^s_{\text{sparse}}$ and the index map $I^s$. The corresponding texture details are obtained by multiplying the sparse index $I^s$ with $V^s$. These details are then weighted by multiplying with the sparsity-related matrix $R^s_{\text{sparse}}$ to obtain the final sparse texture guidance information $T^s$

$$V'_s = V^s \odot I^s, \quad T^s = R^s_{\text{sparse}} \odot V'_s. \tag{22}$$

## E. Loss Function

In our algorithm, there are three loss functions in total. The overall loss function can be interpreted as follows:

$$L = \lambda_1 \times L_{\text{diff}} + \lambda_2 \times L_{\text{pixel}} + \lambda_3 \times L_{\text{per}}. \tag{23}$$

We set the weights of the three loss functions as: $\lambda_1 = 1$, $\lambda_2 = 10^{-3}$, and $\lambda_3 = 10^{-4}$. Below, we specifically introduce the specific meanings of each loss function.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:58:10 UTC from IEEE Xplore. Restrictions apply.

*1) Diffusion Loss:* The objective function of the DM is to minimize the KL divergence between the forward diffusion process $q(I_{t-1}|I_t, I_0, I_{\text{res}})$ and the backward inference $p_\theta(I_{t-1}|I_t)$. Similar to DDPM, we use the following formula to calculate the loss function between the predicted residual and the true residual:

$$L_{\text{res}}(\theta) = \mathbb{E}\left[\lambda_{\text{res}} \| I_{\text{res}} - I^{\theta}_{\text{res}}(I_t, t, I_{\text{in}}) \|^2\right]. \tag{24}$$

The loss function between the predicted noise and the true noise

$$L_{\varepsilon}(\theta) = \mathbb{E}\left[\lambda_{\varepsilon} \| \varepsilon - \varepsilon_\theta(I_t, t, I_{\text{in}}) \|^2\right]. \tag{25}$$

Here, $\lambda_{\text{res}}$, $\lambda_\varepsilon \in [0, 1]$, used for re-weighting. We set $\lambda_{\text{res}} = 1$, $\lambda_\varepsilon = 1$, and the final diffusion loss is

$$L_{\text{diff}}(\theta) = L_{\text{res}}(\theta) + L_{\varepsilon}(\theta). \tag{26}$$

*2) Pixel Loss:* We derive the predicted noise $I^{\theta}_{\varepsilon}$ and residual $I^{\theta}_{\text{res}}$ through the reverse inference process and estimate the original clean image at each time step based on (7). We use the $L1$ pixel loss to calculate the pixel error between SR($\hat{I}_\theta$) and HR($I_0$), with the specific formula being

$$L_{\text{pixel}}(\theta) = \mathbb{E}\left[\ \|\hat{I}^{\theta}_{ijk} - I^{0}_{ijk}\|_1\right]. \tag{27}$$

By introducing pixel loss, we ensure that the SR image conforms to the ground truth.

*3) Perceived Loss:* The core idea of perceptual loss is to compute the difference between two images using a pretrained neural network, thereby enhancing the similarity between the SR image and the HR image in the deep feature space. We use a pretrained VGG network to extract deep features, and the perceptual loss can be interpreted as follows:

$$L_{\text{per}}(\theta) = \mathbb{E}\left[\ \|\phi_{\text{VGG}}\left(\hat{I}^{\theta}_{ijk}\right) - \phi_{\text{VGG}}\left(I^{0}_{ijk}\right)\|^2\right]. \tag{28}$$

Here, $\phi_{\text{VGG}}$ denotes the feature maps of the fourth layer of VGG19. This loss function can minimize the distance between SR and HR in the feature space, thereby enriching the details of the SR image.

## IV. Experimental Details

### A. Dataset

We employ public RS datasets AID, RSD46-WHU, NWPU-RESISC45, and UC MERCED LAND USE to comprehensively evaluate the effectiveness of the proposed method. The AID dataset contains 10 000 images of 30 scene categories, with each image sized 600 × 600. The RSD46-WHU dataset contains 117 000 images of 46 scene categories, with each image sized 256 × 256. The NWPU-RESISC45 dataset contains 31 500 images of 45 scene categories, with each image sized 256 × 256. The UC MERCED LAND USE dataset contains 2100 images of 21 scene categories, with each image sized 256 × 256.

To ensure semantic relevance, Ref images are selected from the same scene category as the input LR image. For each LR input during training and testing, one Ref image is randomly sampled from the corresponding scene category pool in the dataset (e.g., "airport" or "forest" in AID). This leverages the predefined scene labels to guarantee high-level semantic consistency without requiring pixel-level alignment.

To enhance network generalization and reduce computational costs, we crop each image in the AID dataset into four 256 × 256 patches and randomly select one for use. For the RSD46-WHU, NWPU-RESISC45, and UC MERCED LAND USE datasets, we randomly select 100 images per category. Specifically, a total of 21 200 images are selected and divided into training, Ref, and validation sets in a [6:3:1] ratio. Consequently, our training set contains 12 720 images, the Ref set contains 6360 images, and the validation set contains 2120 images. All data are degraded using bicubic interpolation to construct the 4× SR task.

### B. Evaluation Metrics

We use four metrics for SR image quality assessment. Among these, peak signal-to-noise ratio (PSNR) is calculated as the ratio between the maximum signal power and noise power based on mean square error. Structural similarity index measure (SSIM) [73] measures local structural similarity between images through three dimensions: luminance, contrast, and structure. LPIPS [74] is a perceptual similarity metric based on deep feature space, designed to compute weighted distances in the feature space. Fréchet inception distance (FID) [75] evaluates the realism and diversity of generated images by comparing the distribution differences between generated and real images in the Inception-v3 feature space. To evaluate model performance on real-world SR tasks, we utilize the NIQE [76] metric. NIQE is a no-Ref image quality assessment metric that quantifies image naturalness and perceptual quality based on natural scene statistics, making it particularly suitable for real-world SR scenarios where Ref ground-truth images are unavailable.

In summary, PSNR and SSIM focus on pixel-level and structural reconstruction fidelity, whereas LPIPS and FID emphasize human visual perception-based quality assessment. The combined use of these four metrics provides a comprehensive evaluation of algorithmic reconstruction accuracy and perceptual quality. NIQE further provides an assessment of image naturalness under no-Ref conditions.

### C. Implementation Details

Our study focuses on 4× SR. In the final TTRD3 architecture, the number of MFABs in the MFAM is set to [2, 2, 2] to achieve cross-scale feature fusion. For the residual denoising dual diffusion network, we employ a U-Net with an input channel number of $C = 64$, four depth levels, and channel multipliers of [1, 2, 4, 8] at each depth. The TTRD3-2Net network architecture was employed. The noise intensity added is $\bar{\beta}_T = 0.1$. The preset maximum K value is 10.

To train the TTRD3 network, we perform 300k iterations with a batch size of 2. The learning rate is initialized to $1 \times 10^{-4}$, with diffusion loss weight set to 1, pixel loss weight to $1 \times 10^{-3}$, and perceptual loss weight to $1 \times 10^{-4}$. Optimization is conducted using the Adam optimizer with $\beta_1 = 0.9$ and $\beta_2 = 0.99$. The diffusion process uses a total timestep $T = 1000$ and 10 sampling steps.

All comparative methods are retrained on our dataset using their official code implementations and hyperparameters to ensure fair comparison. Experiments are conducted on a single

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:58:10 UTC from IEEE Xplore. Restrictions apply.

**Algorithm 1** Training of TTRD3

**Input** : HR image $I_0$, upsampled LR image $I_{\text{in}}$, Ref image $I_{\text{ref}}$, Downsampled-Upsampled Ref image $I_{\text{refsr}}$, total step $T$

**Initialization:** $\varepsilon \sim \mathcal{N}(0, \delta^2)$, $t \in [0, T]$, $I_{\text{res}} = I_{\text{in}} - I_0$, diffusion network $\psi_\theta$, $\bar{\alpha}_t = \sum_{i=1}^{t} \alpha_i$, $\bar{\beta}_t = \sqrt{\sum_{i=1}^{t} \beta_i^2}$

**repeat**
- $Q = f_{\text{MFAM}}(I_{\text{in}})$, $K = f_{\text{MFAM}}(I_{\text{refsr}})$, $V = f_{\text{MFAM}}(I_{\text{ref}})$ // Multiscale Feature Aggregation
- $T_{\text{guid}} = f_{\text{STTG}}(Q, K, V)$ // Sparse Texture Transfer Guidance
- $I_t = I_0 + \bar{\alpha}_t I_{\text{res}} + \bar{\beta}_t \varepsilon$
- $\varepsilon_\theta, I_{\text{res}}^\theta = \psi_\theta(I_t, I_{\text{in}}, T_{\text{guid}})$ // Predict noise and residual
- $I_0 = I_t - \bar{\alpha}_t I_{\text{res}}^\theta - \bar{\beta}_t \varepsilon_\theta$ // Reconstruct HR image
- $L = \lambda_1 L_{\text{diff}} + \lambda_2 L_{\text{pixel}} + \lambda_3 L_{\text{per}}$ // Loss calculation
- Update $\theta$ via $\nabla_\theta L$ // Gradient descent

**until** *converged*;

**Algorithm 2** Inference of TTRD3

**Input** : Upsampled LR image $I_{\text{in}}$, Ref image $I_{\text{ref}}$, total step $T$, Sampling steps $T_{\text{sample}}$, Downsampled-Upsampled Ref image $I_{\text{refsr}}$

**Output** : SR image $I_{\text{sr}}$

**Initialization:** $\varepsilon \sim \mathcal{N}(0, \delta^2)$, Degrade image: $I_T = I_{\text{in}} + \bar{\beta}_T \varepsilon$, diffusion network $\psi_\theta$

**Inference** :

$Q = f_{\text{MFAM}}(I_{\text{in}})$, $K = f_{\text{MFAM}}(I_{\text{refsr}})$, $V = f_{\text{MFAM}}(I_{\text{ref}})$ // Multiscale Feature Aggregation

$T_{\text{guid}} = f_{\text{STTC}}(Q, K, V)$ // Sparse Texture Transfer Guidance

**for** $t = T$ ***down to*** $1$ ***step*** $\lfloor T/(T_{\text{sample}} - 1) \rfloor$ **do**
- $\alpha_t = \sum_{i=1}^{t} \alpha_i$, $\bar{\beta}_t = \sqrt{\sum_{i=1}^{t} \beta_i^2}$
- $\varepsilon_\theta, I_{\text{res}}^\theta = \psi_\theta(I_t, I_{\text{in}}, T_{\text{guid}})$ // Predict noise and residual
- $I_{\text{prev}} = I_t - (\alpha_t - \alpha_{\text{prev}}) I_{\text{res}}^\theta - (\bar{\beta}_t - \sqrt{\bar{\beta}_{\text{prev}}^2 - \sigma_t^2}) \varepsilon_\theta + \sigma_t \varepsilon$

**end**

$I_{\text{sr}} = \hat{I}_0$ // Predic SR image

NVIDIA RTX 4090 GPU (24GB memory) with the PyTorch framework.

The pseudocode for the training and inference processes of the proposed TTRD3 algorithm can be found in Algorithm 1 and Algorithm 2, respectively.

## V. Experiments and Discussion

### A. Comparison With SOTA Results

This study constructs a comprehensive comparative framework encompassing both SISR and RefSR approaches. For SISR, we selected CNN-based methods RCAN [29], Transformer-based approaches TTST [40], GAN-based models including ESRGAN [42] and LDL [45], as well as DMs including SRDiff [57], SR3 [56], EDiffSR [60], ResShift [58] and StableSR [62]. For RefSR, we included general-purpose methods such as MASA [67] and TTSR [32] along with remote sensing-specific approaches such as RRSGAN [69] and RGTGAN [14]. These methods were chosen as they represent mainstream techniques in the field, ensuring a thorough evaluation to demonstrate our algorithm's effectiveness. To ensure fairness, all methods were trained on our partitioned dataset using the original source codes and hyperparameter settings provided in their respective papers.

*1) Quantitative Comparison:* Table I shows the LPIPS results across 30 scene categories in the AID testing set. The best and second-best LPIPS scores are highlighted in each row. Our TTRD3 achieves superior LPIPS performance in most scenarios, demonstrating robust generalization across diverse RS scenes.

Table II presents the average PSNR, SSIM, LPIPS, and FID metrics as four core evaluation indicators on the testing sets of AID, RSD46-WHU, NWPU-RESISC45, and UC MERCED LAND USE. Comparative analysis with state-of-the-art (SOTA) generative models and RefSR methods reveals that our proposed TTRD3 model achieves optimal performance in both LPIPS and FID, the two key perceptual quality metrics, demonstrating its superior visual generation capability. Notably, while traditional regression-based SISR networks maintain advantages in numerical metrics like PSNR and SSIM, their overly smooth outputs often fail to meet practical perceptual requirements [6]. In contrast, GAN-based and DMs significantly improve texture details, and RefSR techniques further reduce the distribution gap between reconstructed and real images through high-frequency information injection from Ref images. Importantly, TTRD3 not only achieves the best LPIPS and FID scores but also surpasses comparable GAN, DMs, and RefSR methods in traditional metrics like PSNR and SSIM, achieving dual breakthroughs in both perceptual quality and numerical precision.

*2) Qualitative Comparison:* We conducted visual comparisons between our TTRD3 and other models. From the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:58:10 UTC from IEEE Xplore. Restrictions apply.

TABLE I

QUANTITATIVE LPIPS COMPARISON WITH SOTA SR MODELS ON 30 SCENE CATEGORIES OF THE AID TESTING SET. THE BEST LPIPS VALUE IN EACH CATEGORY IS HIGHLIGHTED IN RED, WHILE THE SECOND BEST IS IN BLUE

| Categories | Baseline | Regression-SISR | | GAN-SISR | | Diffusion-SISR | | | | | RefSR | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Bicbic | RCAN | TTST | ESRGAN | LDL | SRDiff | SR3 | EDiffSR | ResShift | StableSR | TTSR | MASA | RRSGAN | RGTGAN | TTRD3 |
| airport | 0.3620 | 0.3041 | 0.2994 | 0.2814 | 0.2783 | 0.2781 | 0.3572 | 0.3844 | 0.2736 | 0.3742 | 0.2654 | 0.3271 | 0.2639 | 0.2599 | 0.2590 |
| bareland | 0.2817 | 0.2554 | 0.2540 | 0.2412 | 0.2325 | 0.2351 | 0.2660 | 0.4299 | 0.3234 | 0.3514 | 0.2282 | 0.2597 | 0.2279 | 0.2284 | 0.2249 |
| baseballfield | 0.2694 | 0.2356 | 0.2340 | 0.2322 | 0.2303 | 0.2295 | 0.2893 | 0.4080 | 0.2465 | 0.3521 | 0.2066 | 0.2506 | 0.2203 | 0.2240 | 0.2024 |
| beach | 0.3529 | 0.3274 | 0.3235 | 0.2893 | 0.2876 | 0.2867 | 0.3402 | 0.3898 | 0.2876 | 0.3684 | 0.2960 | 0.3136 | 0.2779 | 0.2721 | 0.2660 |
| bridge | 0.2923 | 0.2449 | 0.2429 | 0.2304 | 0.2285 | 0.2277 | 0.3174 | 0.3671 | 0.2392 | 0.3457 | 0.2181 | 0.2471 | 0.2148 | 0.2125 | 0.2056 |
| center | 0.3551 | 0.2772 | 0.2724 | 0.2659 | 0.2652 | 0.2638 | 0.3764 | 0.3573 | 0.2494 | 0.3703 | 0.2476 | 0.3114 | 0.2524 | 0.2455 | 0.2437 |
| church | 0.3942 | 0.3335 | 0.3314 | 0.3070 | 0.3033 | 0.3034 | 0.3912 | 0.3626 | 0.3024 | 0.3803 | 0.2906 | 0.3582 | 0.2877 | 0.2839 | 0.2796 |
| commercial | 0.3755 | 0.3199 | 0.3173 | 0.2982 | 0.3003 | 0.2975 | 0.3886 | 0.3750 | 0.2931 | 0.3852 | 0.2774 | 0.3569 | 0.2823 | 0.2753 | 0.2747 |
| denseresidential | 0.4299 | 0.3590 | 0.3533 | 0.3066 | 0.3114 | 0.3072 | 0.4106 | 0.3570 | 0.3149 | 0.3872 | 0.2885 | 0.3957 | 0.2898 | 0.2805 | 0.2802 |
| desert | 0.2865 | 0.2522 | 0.2488 | 0.2281 | 0.2257 | 0.2251 | 0.3003 | 0.4145 | 0.2617 | 0.3608 | 0.2297 | 0.2495 | 0.2201 | 0.2199 | 0.2170 |
| farmland | 0.3430 | 0.2921 | 0.2936 | 0.2745 | 0.2755 | 0.2732 | 0.3645 | 0.4246 | 0.2820 | 0.3980 | 0.2690 | 0.3051 | 0.2640 | 0.2646 | 0.2555 |
| forest | 0.4092 | 0.4028 | 0.4027 | 0.3196 | 0.3648 | 0.3404 | 0.3482 | 0.3803 | 0.3279 | 0.3677 | 0.3303 | 0.3546 | 0.3073 | 0.3099 | 0.2919 |
| industrial | 0.3369 | 0.2669 | 0.2606 | 0.2529 | 0.2477 | 0.2485 | 0.3486 | 0.3165 | 0.2360 | 0.3360 | 0.2277 | 0.3035 | 0.2338 | 0.2287 | 0.2281 |
| meadow | 0.3489 | 0.3362 | 0.3338 | 0.3094 | 0.3068 | 0.3063 | 0.3283 | 0.4319 | 0.4137 | 0.3835 | 0.3104 | 0.3134 | 0.2896 | 0.2976 | 0.2835 |
| mediumresidential | 0.3749 | 0.3339 | 0.3284 | 0.2771 | 0.2845 | 0.2790 | 0.3379 | 0.3261 | 0.2726 | 0.3354 | 0.2630 | 0.3267 | 0.2563 | 0.2558 | 0.2526 |
| mountain | 0.3680 | 0.3514 | 0.3487 | 0.3238 | 0.3202 | 0.3202 | 0.3614 | 0.4183 | 0.3132 | 0.3933 | 0.3105 | 0.3464 | 0.3080 | 0.3067 | 0.3042 |
| park | 0.3496 | 0.3094 | 0.3074 | 0.2896 | 0.2949 | 0.2905 | 0.3537 | 0.3829 | 0.2792 | 0.3717 | 0.2669 | 0.3300 | 0.2724 | 0.2700 | 0.2673 |
| parking | 0.3094 | 0.2298 | 0.2264 | 0.2411 | 0.2371 | 0.2373 | 0.3472 | 0.3018 | 0.2305 | 0.3279 | 0.2025 | 0.2766 | 0.2218 | 0.2184 | 0.2011 |
| playground | 0.3098 | 0.2490 | 0.2482 | 0.2308 | 0.2277 | 0.2275 | 0.3090 | 0.3562 | 0.2497 | 0.3360 | 0.2209 | 0.2571 | 0.2137 | 0.2135 | 0.2094 |
| pond | 0.2750 | 0.2537 | 0.2583 | 0.2332 | 0.2402 | 0.2349 | 0.3385 | 0.3940 | 0.2471 | 0.3697 | 0.2271 | 0.2394 | 0.2288 | 0.2231 | 0.2102 |
| port | 0.3026 | 0.2611 | 0.2584 | 0.2447 | 0.2435 | 0.2423 | 0.3475 | 0.3690 | 0.2395 | 0.3617 | 0.2291 | 0.2756 | 0.2333 | 0.2301 | 0.2209 |
| railwaystation | 0.3812 | 0.3074 | 0.3012 | 0.2867 | 0.2802 | 0.2817 | 0.3849 | 0.3521 | 0.2743 | 0.3719 | 0.2698 | 0.3398 | 0.2645 | 0.2580 | 0.2513 |
| resort | 0.3209 | 0.2753 | 0.2711 | 0.2574 | 0.2549 | 0.2544 | 0.3238 | 0.3345 | 0.2472 | 0.3326 | 0.2340 | 0.2957 | 0.2423 | 0.2379 | 0.2299 |
| river | 0.3481 | 0.3204 | 0.3169 | 0.2785 | 0.2847 | 0.2798 | 0.3357 | 0.3716 | 0.2734 | 0.3571 | 0.2768 | 0.3151 | 0.2647 | 0.2617 | 0.2550 |
| school | 0.3915 | 0.3330 | 0.3286 | 0.2981 | 0.3007 | 0.2976 | 0.3846 | 0.3642 | 0.3009 | 0.3778 | 0.2746 | 0.3641 | 0.2827 | 0.2762 | 0.2719 |
| sparseresidential | 0.4050 | 0.3919 | 0.3880 | 0.3073 | 0.3225 | 0.3131 | 0.3327 | 0.3367 | 0.3059 | 0.3381 | 0.3113 | 0.3624 | 0.2849 | 0.2811 | 0.2797 |
| square | 0.2921 | 0.2371 | 0.2337 | 0.2182 | 0.2170 | 0.2158 | 0.2975 | 0.3183 | 0.2117 | 0.3113 | 0.1968 | 0.2590 | 0.2041 | 0.1999 | 0.1996 |
| stadium | 0.3133 | 0.2492 | 0.2453 | 0.2449 | 0.2407 | 0.2410 | 0.3224 | 0.3331 | 0.2376 | 0.3312 | 0.2266 | 0.2710 | 0.2270 | 0.2264 | 0.2214 |
| storagetanks | 0.3765 | 0.3247 | 0.3229 | 0.2862 | 0.2850 | 0.2838 | 0.3555 | 0.3493 | 0.2758 | 0.3558 | 0.2713 | 0.3394 | 0.2685 | 0.2626 | 0.2641 |
| viaduct | 0.3703 | 0.3044 | 0.3057 | 0.2772 | 0.2803 | 0.2770 | 0.3465 | 0.3411 | 0.2611 | 0.3472 | 0.2622 | 0.3199 | 0.2596 | 0.2525 | 0.2493 |
| AVERAGE | 0.3442 | 0.2980 | 0.2952 | 0.2711 | 0.2724 | 0.2699 | 0.3435 | 0.3683 | 0.2757 | 0.3593 | 0.2576 | 0.3088 | 0.2555 | 0.2526 | 0.2467 |

TABLE II

QUANTITATIVE COMPARISONS WERE CONDUCTED ON THE AID, RSD46-WHU, NWPU-RESISC45, AND UC MERCED LAND USE TESTING SETS WITH SOTA SR MODELS. IN THE COMPARISONS WITH GAN-BASED, DIFFUSION-BASED SISR METHODS AND REFSR APPROACHES, THE BEST METRICS ARE HIGHLIGHTED IN RED WHILE THE SECOND-BEST METRICS ARE MARKED IN BLUE

| Dataset | Metrics | Baseline | Regression-SISR | | GAN-SISR | | Diffusion-SISR | | | | | RefSR | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Bicbic | RCAN | TTST | ESRGAN | LDL | SRDiff | SR3 | EDiffSR | ResShift | StableSR | TTSR | MASA | RRSGAN | RGTGAN | TTRD3 |
| AID | PSNR(↑) | 27.9757 | 30.1317 | 30.1627 | 27.4881 | 27.7927 | 27.9496 | 26.4684 | 25.2836 | 28.1733 | 24.5830 | 28.7728 | 28.1082 | 28.4206 | 28.5345 | 28.8432 |
| | SSIM(↑) | 0.7298 | 0.8156 | 0.8172 | 0.7377 | 0.7533 | 0.7424 | 0.6921 | 0.5946 | 0.7498 | 0.6004 | 0.7805 | 0.7374 | 0.7517 | 0.7593 | 0.7808 |
| | LPIPS(↓) | 0.3442 | 0.2980 | 0.2952 | 0.2711 | 0.2724 | 0.2699 | 0.3435 | 0.3683 | 0.2757 | 0.3593 | 0.2576 | 0.3088 | 0.2555 | 0.2526 | 0.2467 |
| | FID(↓) | 90.7452 | 84.8817 | 83.0137 | 40.4819 | 39.9917 | 39.1780 | 77.2483 | 59.0955 | 59.8146 | 49.2168 | 57.9357 | 58.6754 | 38.8088 | 40.7482 | 32.7252 |
| RSD46-WHU | PSNR(↑) | 27.8750 | 29.7836 | 29.7858 | 26.9006 | 28.4559 | 27.1357 | 26.2499 | 24.8892 | 28.1425 | 24.5462 | 28.6293 | 27.6096 | 28.6126 | 28.7570 | 28.8605 |
| | SSIM(↑) | 0.7267 | 0.8069 | 0.8082 | 0.7105 | 0.7615 | 0.7078 | 0.6801 | 0.5783 | 0.7561 | 0.6377 | 0.7686 | 0.7244 | 0.7556 | 0.7593 | 0.7643 |
| | LPIPS(↓) | 0.3392 | 0.2998 | 0.2991 | 0.2758 | 0.2526 | 0.2788 | 0.3557 | 0.3888 | 0.2825 | 0.3696 | 0.2519 | 0.2597 | 0.2480 | 0.2518 | 0.2418 |
| | FID(↓) | 115.8997 | 111.7993 | 111.4668 | 60.4716 | 55.9034 | 67.9859 | 107.7770 | 87.5350 | 95.2061 | 83.9301 | 75.7809 | 60.1659 | 55.3338 | 57.7431 | 49.9658 |
| NWPU-RESISC45 | PSNR(↑) | 26.4655 | 28.2316 | 28.2388 | 25.4766 | 27.0315 | 25.9749 | 25.0475 | 23.9600 | 26.9844 | 23.7345 | 27.0107 | 25.8506 | 27.0975 | 27.2657 | 27.3974 |
| | SSIM(↑) | 0.6917 | 0.7801 | 0.7798 | 0.6769 | 0.7344 | 0.6762 | 0.6428 | 0.5607 | 0.7333 | 0.5579 | 0.7477 | 0.6840 | 0.7283 | 0.7328 | 0.7486 |
| | LPIPS(↓) | 0.3630 | 0.3237 | 0.3235 | 0.2859 | 0.2617 | 0.2972 | 0.3753 | 0.3836 | 0.2980 | 0.3801 | 0.2748 | 0.2744 | 0.2613 | 0.2663 | 0.2589 |
| | FID(↓) | 128.2654 | 133.8027 | 131.8874 | 61.4072 | 62.1782 | 69.1380 | 115.5776 | 84.3360 | 113.0756 | 72.9311 | 90.7084 | 66.7040 | 60.8706 | 61.5629 | 51.3766 |
| UC MERCED LAND USE | PSNR(↑) | 25.2852 | 27.4854 | 27.5623 | 24.7853 | 26.2436 | 25.2076 | 24.5118 | 23.3858 | 26.1908 | 23.0987 | 26.6160 | 25.2230 | 26.4262 | 26.5564 | 26.7812 |
| | SSIM(↑) | 0.6816 | 0.7759 | 0.7779 | 0.6798 | 0.7302 | 0.6918 | 0.6530 | 0.5766 | 0.7329 | 0.6097 | 0.7512 | 0.6842 | 0.7255 | 0.7292 | 0.7580 |
| | LPIPS(↓) | 0.3921 | 0.3297 | 0.3282 | 0.3010 | 0.2701 | 0.2935 | 0.3736 | 0.3708 | 0.3033 | 0.3392 | 0.2788 | 0.2811 | 0.2707 | 0.2745 | 0.2662 |
| | FID(↓) | 119.3554 | 114.4566 | 111.6235 | 66.6626 | 56.5443 | 67.8739 | 110.9436 | 88.4251 | 89.9129 | 70.2520 | 67.7632 | 57.1048 | 54.5856 | 56.2610 | 47.2872 |

comparative results shown in Fig. 7 of the AID dataset, it can be observed that our proposed TTRD3 model achieves the best visual realism and accuracy. Among SISR methods, regression-based approaches, including RCAN and TTST produce overly smooth results due to their pixelwise optimization characteristics. While GAN-based networks ESRGAN and LDL enhance textures, they generate repetitive artifacts. DMs, including SRDiff, SR3, EDiffSR, ResShift, and StableSR, face challenges in balancing authenticity and accuracy. Although RefSR-based methods can supplement additional high-frequency information, the CNN-based MASA method remains limited by its feature representation capability, still suffering from over-smoothing issues. While GAN-based approaches like TTSR, RRSGAN, and RGTGAN alleviate over-smoothing, they tend to converge to local optima during training, generating repetitive artifacts in forested scenes. Our TTRD3, which recovers images through dual diffusion of residuals and noise, can generate highly realistic details while maintaining excellent accuracy, precisely restoring high-frequency textures like zebra crossings in urban scenes while preserving the geometric consistency of vehicle contours and vegetation details. This dual-constraint strategy achieves a breakthrough balance between generation quality and structural fidelity.

Fig. 8 presents extended validation on testing sets of RSD46-WHU, NWPU-RESISC45, and UC MERCED LAND USE, further revealing model characteristics. Among SISR methods, regression-based RCAN and TTST demonstrate

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:58:10 UTC from IEEE Xplore. Restrictions apply.




Fig. 7. Visual comparison of different networks on the AID dataset for the 4× SR task, with corresponding LPIPS metrics labeled below each SR image. Zoomed-in view for better details.

basic reconstruction capability for regular structures like building edges but appear visually blurry overall. While GAN-based and DMs generate rich details, they exhibit deformation offsets in structured targets like container arrangements and parking lots. RefSR methods generally outperform SISR in LPIPS metrics due to supplementary information, yet still cannot avoid low-frequency smoothing. Interestingly, in aircraft runway scenarios of NWPU-RESISC45, GAN-based networks show good LPIPS performance despite severe artifacts, indicating their tendency to reach local optima due to perceptual loss. In contrast, our diffusion-based TTRD3 simultaneously achieves vivid high-frequency details and precise geometric structures across various scenarios, demonstrating significantly better recovery of complex high-frequency targets like container edges and airport runways compared with other methods. This confirms the theoretical advantages of our dual-path diffusion mechanism in texture generation and structural constraints.

*3) Real-World Comparison:* We evaluate the performance of TTRD3 on real-world SR tasks, which involve directly super-resolving pristine HR images without simulated degradation processes. As shown in Table III, a quantitative comparison of NIQE scores between TTRD3 and other SOTA methods reveals that despite no explicit modifications for real-world scenarios, TTRD3 achieves competitive performance second only to StableSR. This demonstrates the model's robust generative capability in effectively restoring high-frequency details such as edge textures.

Notably, StableSR, which leverages a powerful pre-trained generative prior, leads in perceptual quality metrics,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:58:10 UTC from IEEE Xplore. Restrictions apply.

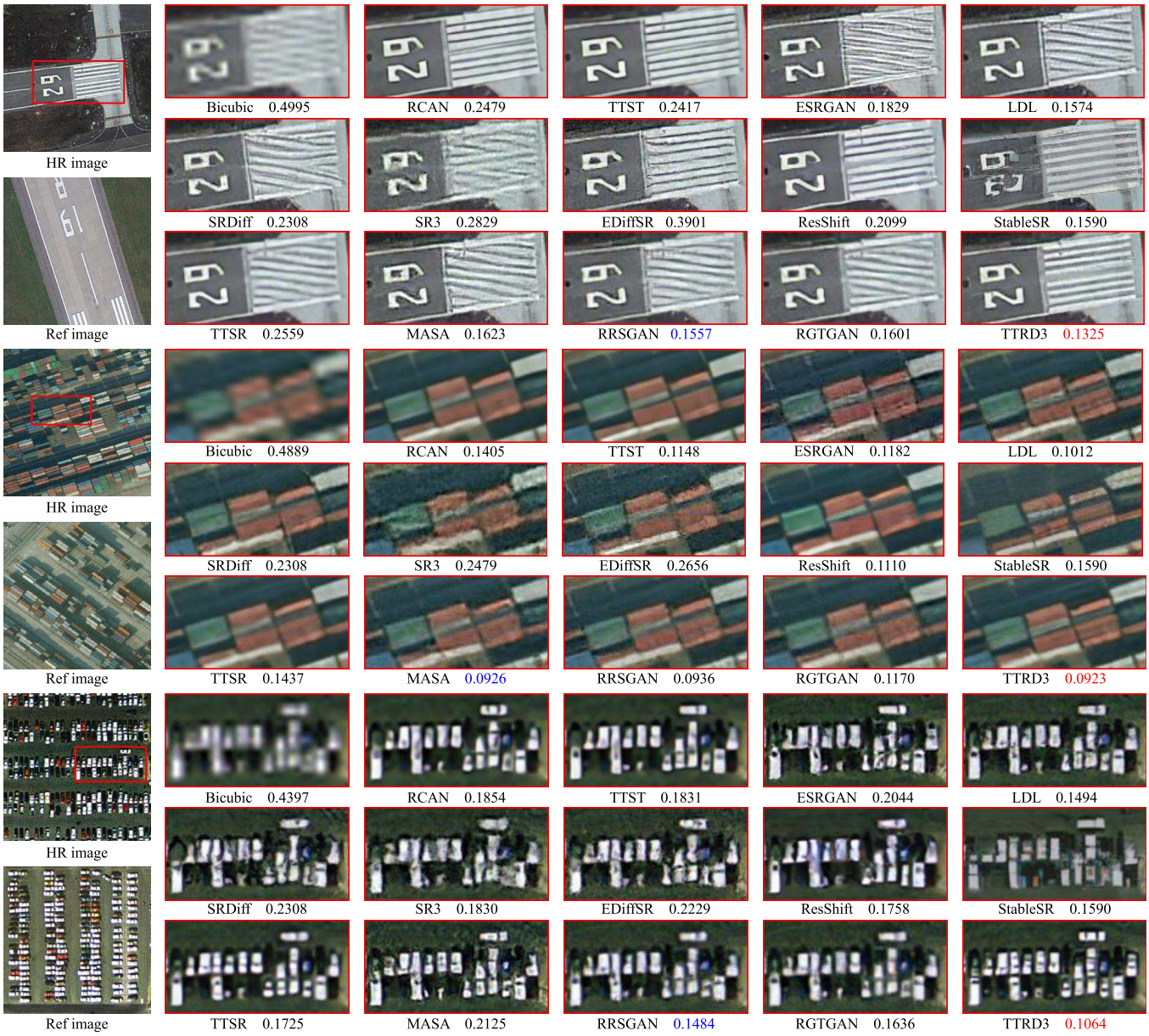



Fig. 8. Three sets of visual comparisons demonstrate the network performance on RSD46-WHU, NWPU-RESISC45, and UC MERCED LAND USE datasets for the 4× SR task, respectively, with LPIPS evaluation scores clearly indicated below each super-resolved image. Zoomed-in view for better details.

TABLE III

QUANTITATIVE NIQE COMPARISON WITH SOTA SR MODELS ON AID WITH REAL-WORLD DEGRADATIONS. THE BEST NIQE VALUE IN EACH CATEGORY IS HIGHLIGHTED IN RED, WHILE THE SECOND BEST IS IN BLUE

| Metrics | Bicubic | TTST | ESRGAN | LDL | SR3 | SRDiff | EDiffSR | ResShift | StableSR | TTSR | MASA | RRSGAN | RGTGAN | TTRD3 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| NIQE(↓) | 8.52 | 8.21 | 7.24 | 6.79 | 7.61 | 6.76 | 6.12 | 6.97 | 5.12 | 6.77 | 6.83 | 6.71 | 6.68 | 5.78 |

highlighting a tradeoff between leveraging large-scale pre-trained models and designing specialized architectures like ours. Qualitatively, as shown in Fig. 9, TTRD3 excels in reconstructing geometrically accurate structures, while other methods either introduce artifacts or oversmooth details. These findings confirm that TTRD3 establishes a robust balance between perceptual quality and geometric fidelity.

### B. Ablation Study on Diffusion Parameters

To comprehensively analyze the impact of key diffusion parameters on the SR performance of TTRD3, we conduct ablation studies on three aspects: the number of sampling steps $T$, the maximum noise intensity $\bar{\beta}_T$, and the scheduling strategies for residual and noise coefficients ($\alpha_t$ and $\beta_t^2$). All experiments are performed on the AID testing set.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:58:10 UTC from IEEE Xplore. Restrictions apply.




Fig. 9. 4× visual comparisons with SOTA SR models on AID with real-world degradations. Zoomed-in view for a better view and better details.

TABLE IV
PERFORMANCE COMPARISON OF TTRD3 ON THE AID TESTING SET UNDER DIFFERENT CONFIGURATIONS

| Schedules | $T$ | $\bar{\beta}_T^2$ | PSNR↑ | SSIM↑ | LPIPS↓ | FID↓ |
|---|---|---|---|---|---|---|
| $\alpha_t$ (linearly decreasing), $\beta_t^2$ (linearly increasing) | 10 | 0.1 | 28.8418 | 0.7801 | 0.2473 | 32.7873 |
| | | 0.3 | 28.8432 | 0.7808 | 0.2467 | 32.7252 |
| | | 0.5 | 28.8628 | 0.7886 | 0.2613 | 34.5713 |
| | | 0.7 | 28.9307 | 0.7897 | 0.2688 | 36.3682 |
| | | 1 | 29.0465 | 0.7919 | 0.2723 | 38.2344 |
| | 5 | 0.3 | 29.3393 | 0.7999 | 0.2535 | 40.0347 |
| | 10 | | 28.8432 | 0.7808 | 0.2467 | 32.7252 |
| | 15 | | 28.6528 | 0.7745 | 0.2472 | 32.3950 |
| | 20 | | 28.5171 | 0.7695 | 0.2465 | 31.8126 |
| | 30 | | 28.3515 | 0.7547 | 0.2411 | 30.2837 |
| $\alpha_t$ (mean), $\beta_t^2$ (mean) | 10 | 0.3 | 27.9735 | 0.7413 | 0.2732 | 44.4793 |
| $\alpha_t$ (linearly increasing), $\beta_t^2$ (linearly increasing) | | | 28.0382 | 0.7389 | 0.2697 | 41.5682 |
| $\alpha_t$ (linearly decreasing), $\beta_t^2$ (linearly decreasing) | | | 28.1380 | 0.7577 | 0.2671 | 39.1789 |
| $\alpha_t$ (linearly increasing), $\beta_t^2$ (linearly decreasing) | | | 27.7382 | 0.7189 | 0.2798 | 47.5682 |

*1) Sampling Steps $T$:* As shown in Table IV, varying the number of sampling steps $T$ significantly affects both the efficiency and performance of TTRD3. With fewer steps (e.g., $T = 5$), the model achieves faster inference but suffers from reduced perceptual quality. As $T$ increases, the perceptual metrics (LPIPS and FID) improve consistently, indicating that more iterative refinements lead to more realistic textures. However, beyond $T = 15$, the gains diminish while computational cost increases linearly. We find that $T = 10$ offers a favorable balance between performance and efficiency for most scenarios.

*2) Noise Intensity $\bar{\beta}_T$:* The hyper-parameter $\bar{\beta}_T$ controls the maximum noise level in the diffusion process. We evaluate three values: $\bar{\beta}_T$=0.1,0.3,0.5,0.7, and 1.0. Quantitative results in Table IV indicate that an intermediate noise intensity ($\bar{\beta}_T = 0.3$) achieves the optimal balance, yielding the best scores across PSNR, SSIM, LPIPS, and FID metrics. However, visual analysis reveals a critical tradeoff. While a higher $\bar{\beta}_T$ (e.g., 1.0) can enhance textural appearance, it introduces obvious structural hallucinations, such as erroneous road trajectories and distorted vehicles, as shown in Fig. 10(a). Conversely, a lower $\bar{\beta}_T$ (0.1) over-constrains the generation, producing perceptually unsatisfactory results. Thus, $\bar{\beta}_T = 0.3$ is selected as the default, as it provides sufficient generative freedom for realistic textures while the residual diffusion pathway effectively preserves geometric fidelity and prevents such semantic distortions.

TABLE V
ABLATION STUDY RESULTS OF EACH MODULE IN TTRD3 ON THE AID TESTING SET

| Methods | DDPM | RDDM | STTG | MFAM | PSNR↑ | SSIM↑ | LPIPS↓ | FID↓ |
|---|---|---|---|---|---|---|---|---|
| Baseline | ✓ | × | × | × | 26.4684 | 0.6921 | 0.3435 | 72.2483 |
| Model-1 | × | ✓ | × | × | 28.2994 | 0.7497 | 0.2675 | 36.4484 |
| Model-2 | × | ✓ | ✓ | × | 28.7099 | 0.7714 | 0.2501 | 33.0630 |
| TTRD3 | × | ✓ | ✓ | ✓ | 28.8432 | 0.7808 | 0.2467 | 32.7252 |

*3) Scheduling Strategies for $\alpha_t$ and $\beta_t^2$:* We comprehensively evaluate five distinct scheduling strategies for the residual ($\alpha_t$) and noise ($\beta_t^2$) diffusion coefficients. The quantitative results, summarized in Table IV, demonstrate that the choice of schedule significantly impacts reconstruction performance.

The strategy combining a linearly decreasing $\alpha_t$ with a linearly increasing $\beta_t^2$ achieves the best performance across all metrics. This design is empirically optimal as it ensures strong structural guidance from the residual signal at the beginning of the reverse process, which is then progressively refined by an increasing level of stochastic noise to generate realistic textures.

Conversely, the strategy of linearly increasing $\alpha_t$ and linearly decreasing $\beta_t^2$ yields the worst results. This counterintuitive schedule weakens the deterministic structural guidance over time while simultaneously suppressing the stochastic refinement, leading to a significant loss of both fidelity and perceptual quality. The visual superiority of the proposed strategy and the artifacts generated by suboptimal schedules are clearly demonstrated in the qualitative comparisons provided in Fig. 10(b).

### C. Ablation Study on Model Architecture

In this section, we conduct extensive experiments to validate the effectiveness of each component in our TTRD3 framework. As shown in Table V.

*1) Effectiveness of the RDDM:* To validate the effectiveness of the RDDM in SR tasks, we remove the MFAM and STTG, using only the LR image and the degraded image $I_T$ (via channel concatenation) as network inputs. Experimental results show that Model-1 significantly outperforms the baseline

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:58:10 UTC from IEEE Xplore. Restrictions apply.

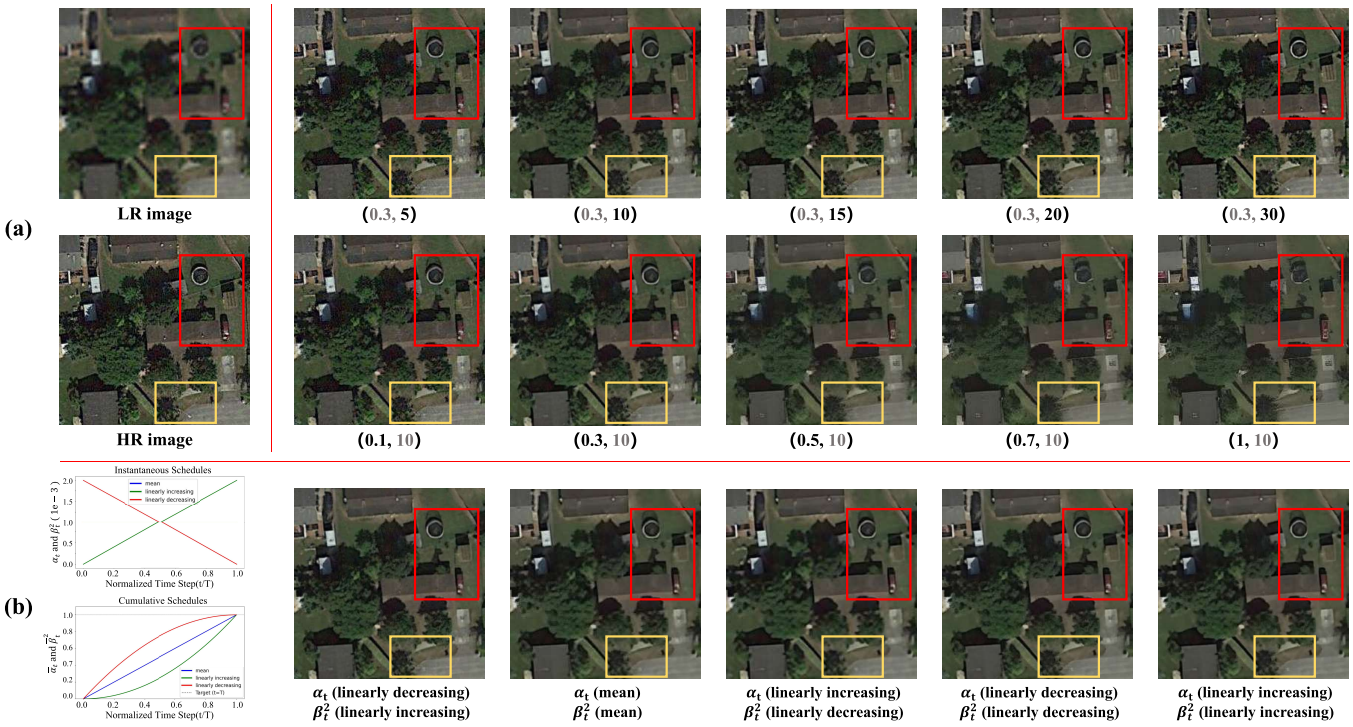



Fig. 10. Qualitative analysis of TTRD3 under different parameter configurations. (a) Visual comparisons of SR results under different combinations of ($\bar{\beta}_T^2$, T). For example, "(0.3, 10)" represents the recovered result with $\bar{\beta}_T^2 = 0.3$ and $T = 10$. (b) Visual comparisons of results generated by different scheduling strategies for the coefficients $\alpha_t$ and $\beta_t^2$. Please zoomed-in view for a better view of the textural and structural details.

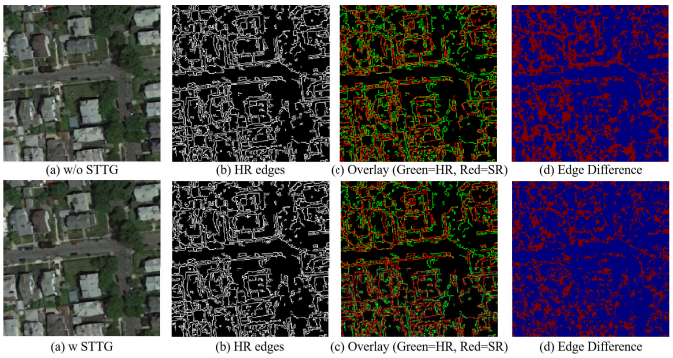



Fig. 11. Visualization of ablation comparisons for the STTG module. (a) SR results with and without the STTG module, (b) boundary maps of HR images obtained using the Canny operator, (c) overlay comparison between HR and SR boundaries, and (d) visualized boundary discrepancies.

model relying solely on noise diffusion across all metrics (PSNR: + 1.8310 dB, SSIM: + 0.0576, LPIPS: −0.0760, and FID: −35.7999). This confirms that residual information enhances reconstruction fidelity, while noise diffusion optimizes generation quality.

*2) Effectiveness of the STTG:* To evaluate the contribution of STTG, we integrate it into the RDDM framework and replace the MFAB in MFAM with standard residual blocks, constructing Model-2. Comparative experiments demonstrate that Model-2 achieves superior performance over Model-1 across all metrics (PSNR: + 0.4105 dB, SSIM: + 0.0217, LPIPS: −0.0174, and FID: −3.3854). Fig. 11 demonstrates the ablation study results of the STTG. This proves that STTG effectively extracts sparse texture guidance from Ref images and significantly enhances the SR generation performance of the dual DM.

*3) Effectiveness of the MFAM:* By comparing the full TTRD3 framework with Model-2, we observe that TTRD3 achieves significant advantages in PSNR, SSIM, LPIPS, and FID (PSNR: + 0.1333 dB, SSIM: + 0.0094, LPIPS: −0.0034, and FID: −0.3378). This validates the critical role of MFAB in extracting multiscale feature information from RS images, providing essential supplementary information for STTG's sparse texture extraction, thereby further improving SR performance.

### D. Analysis of the Top-K Selection Mechanism in STTG

*1) Tradeoff Analysis Between Noise Suppression and Detail Preservation:* To investigate the tradeoff relationship of the Top-$K$ selection mechanism between noise suppression and detail preservation, this study designed comparative experiments with and without the Top-$K$ mechanism. From the SR results, the SR image without the Top-$K$ mechanism exhibited more artifact information, failing to adequately restore building edges and authentic tree textures [see Fig. 12(c)]. In contrast, the SR image with the Top-$K$ mechanism achieved clear reconstruction of buildings and trees [see Fig. 12(d)].

*2) Validation of the Necessity of Dynamic Top-K Selection:* To investigate the effectiveness of the dynamically adjusted Top-$K$ feature selection mechanism, we compared the performance of fixed-$K$ and dynamic-$K$ strategies in complex scenarios (commercial) and homogeneous scenarios (bareland). Experimental results demonstrate that in commercial

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:58:10 UTC from IEEE Xplore. Restrictions apply.

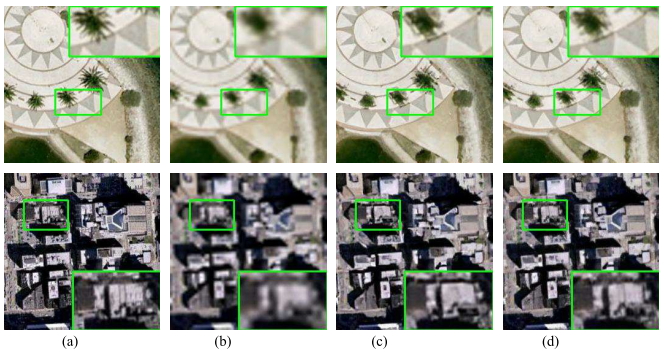



Fig. 12. Ablation visualization comparison of Top-$K$ selection mechanism. (a) HR. (b) Bicubic. (c) w/o Top-K. (d) w/ Top-K.

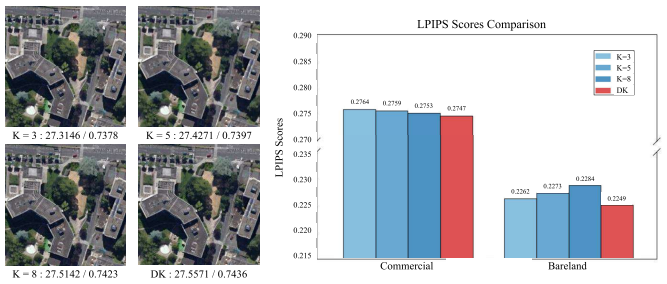



Fig. 13. Performance comparison between fixed $K$-value and dynamic $K$-value in complex scenarios versus uniform scenarios. Subscript numbers denote the PSNR/SSIM score for this example image.

TABLE VI

COMPARISON EXPERIMENTAL RESULTS OF REF IMAGES WITH DIFFERENT RELEVANCE LEVELS IN THE AID TESTING DATASET

| Reference Image | PSNR↑ | SSIM↑ | LPIPS↓ | FID↓ |
|---|---|---|---|---|
| Relevant | 28.8432 | 0.7808 | 0.2467 | 32.7252 |
| Irrelevant | 28.4188 | 0.7582 | 0.2579 | 34.2071 |
| Noise | 28.3043 | 0.7510 | 0.2685 | 35.8484 |

scenarios with rich structural features, higher K-values effectively integrate more detailed information from Ref images, significantly improving the LPIPS metrics of reconstructed images. Conversely, in homogeneous scenarios like bareland with uniform textures, higher $K$-values tend to introduce additional noise, leading to degraded LPIPS performance. Fig. 13 further validates the scenario adaptability of our method: reconstruction quality in commercial areas matches the performance of the fixed $K = 8$ strategy, while in bareland regions it approaches the optimal fixed $K = 3$ strategy. This complexity-aware dynamic adjustment mechanism effectively addresses the performance fluctuations of fixed-K strategies in cross-scenario applications, demonstrating robust adaptability for Ref-based image SR tasks.

### *E. Semantic Relevance Analysis of Ref Images*

To investigate the impact of Ref image semantic relevance on the SR performance of TTRD3, this experiment covers three typical scenarios: strongly relevant Ref (HR images from the same scene), weakly relevant Ref (HR images from different scenes), and noise Ref (Gaussian noise). The

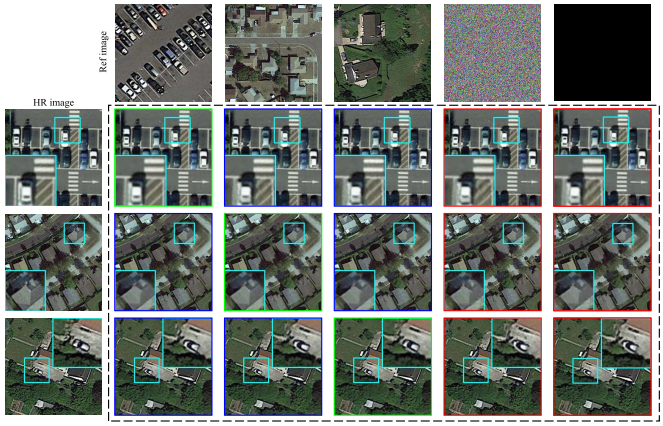



Fig. 14. Comparison of the impact of different Ref images on SR results, where the SR images generated from strongly relevant Ref images are marked with green boxes, the SR images generated from weakly relevant Ref images are marked with blue boxes, and the SR images generated from noise Ref image and black Ref image are marked with red boxes.

quantitative analysis results in Table VI show that when using strongly relevant Ref images, the model achieves optimal performance, outperforming the metrics of weakly relevant Ref images (PSNR: + 0.4244 dB, SSIM: + 0.0226, LPIPS: −0.0112, and FID: −1.4819), which validates the effectiveness of the multiscale texture transfer mechanism in extracting and fusing semantically matched features.

Notably, even under weakly relevant Ref conditions, the model maintains a performance advantage compared with the noise Ref (PSNR: + 0.1145 dB, SSIM: + 0.0072, LPIPS: −0.0106, and FID: −1.6413), indicating that cross-scene feature transfer possesses a certain degree of generalization capability. When the Ref image degenerates into Gaussian noise, the model reverts to RDDM, and its performance metrics are highly consistent with those of the no-Ref baseline model Model-1. This phenomenon confirms the strong robustness characteristics of TTRD3.

Fig. 14 demonstrates the impact of different levels of semantic relevance in Ref images on SR results. When the Ref image is strongly semantically relevant (same scene), the SR results exhibit optimal visual quality: car edges are sharp, zebra crossing textures are clear, and building structures are accurately aligned. When the Ref image is weakly semantically relevant (different scene), the reconstruction quality slightly decreases, slightly blurred edges and misalignment. Even when the Ref image is pure noise, the model maintains basic reconstruction capabilities, further validating its robustness.

To intuitively validate the guiding effect of Ref images on texture transfer, this study conducted a visualization analysis of the multiscale output features of the STTG module, with the results shown in Fig. 15. By comparing the feature response maps of strongly relevant, weakly relevant, and noise Ref images, significant differences can be observed: the feature maps of strongly relevant Ref images exhibit clear structured texture patterns, with edge responses highly consistent with the target regions; although weakly relevant Ref images can activate some features, their response patterns show obvious spatial dispersion; while the feature maps of noise Ref images

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:58:10 UTC from IEEE Xplore. Restrictions apply.

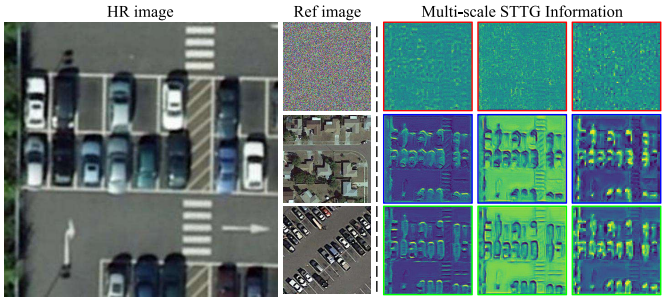



Fig. 15. Visualization of STTG output features for different Ref images, where features from strongly relevant Ref images are marked with green boxes, features from weakly relevant Ref images are marked with blue boxes, and features from noise Ref images are marked with red boxes.

TABLE VII
COMPARISON OF METRICS FOR THREE RESIDUAL DENOISING PREDICTION NETWORKS ON THE AID TESTING SET

| Methods | PSNR↑ | SSIM↑ | LPIPS↓ | FID↓ | Params | Inference time |
|---|---|---|---|---|---|---|
| TTRD3-1Net-A | 28.6946 | 0.7698 | 0.2511 | 36.5526 | 54.98M | 0.2558s |
| TTRD3-1Net-B | 28.8538 | 0.7819 | 0.2487 | 34.0643 | 80.48M | 0.3932s |
| TTRD3-2Net | 28.8432 | 0.7808 | 0.2467 | 32.7252 | 110.11M | 0.5365s |

display uniformly distributed low-frequency noise, without revealing any effective texture structures. This visualization result confirms the critical role of Ref images in the texture transfer process at the feature level, providing intuitive evidence for the feasibility of the multiscale texture guidance mechanism.

### F. Analysis of the Residual Denoising U-Net Structure

In this study, an in-depth exploration of the U-Net architecture within the RDDM framework was conducted. Since RDDM generates SR images by jointly predicting residual information and noise information, its core prediction network can adopt either a single U-Net or dual U-Net architecture. For the single U-Net implementation, two distinct approaches were validated: the first is a fully shared architecture (TTRD3-1Net-A), where residual and noise predictions share the complete encoder–decoder path, with the network output layer expanded to six channels (the first three channels predict residuals, and the latter three predict noise). The second approach is an encoder-shared architecture (TTRD3-1Net-B), which uses a shared encoder for feature extraction but employs independent decoders to separately predict residuals and noise. As a comparative baseline, the dual U-Net independent architecture (TTRD3-2Net) utilizes completely separate networks for residual and noise prediction, as illustrated in Fig. 16.

Comparative experiments conducted on an NVIDIA RTX 4090 GPU (as detailed in Table VII) revealed that the single U-Net approach significantly reduces computational costs at the expense of residual-noise decoupling. TTRD3-1Net-A and TTRD3-1Net-B reduced parameter counts by 50.07% and 26.89%, respectively, compared with TTRD3-2Net, while inference times per image decreased by 52.32% and 26.93%. However, parameter compression and feature coupling led to a decline in SR metrics: TTRD3-1Net-A exhibited the most significant performance degradation due to fully shared encoder–decoder layers. TTRD3-1Net-B partially preserved decoupling through independent decoders, achieving superior PSNR and SSIM metrics compared with the dual-network architecture. Although TTRD3-2Net achieved optimal LPIPS and FID scores, it required higher training costs and inference latency. Further analysis indicated that TTRD3-1Net-B achieves a relative balance between efficiency and performance, making it suitable for deployment in computationally constrained scenarios, while TTRD3-2Net is better suited for applications demanding rigorous reconstruction quality. These experimental results provide practical guidance for model selection based on hardware conditions and task objectives.

TABLE VIII
COMPARISON OF METRIC PERFORMANCE UNDER DIFFERENT TIME STEPS ON THE AID TESTING SET

| Methods | Timesteps | Inference time | PSNR↑ | SSIM↑ | LPIPS↓ | FID↓ |
|---|---|---|---|---|---|---|
| TTRD3-1Net-A | 5 | 0.1141 | 29.1881 | 0.7868 | 0.2562 | 42.3455 |
| | 10 | 0.2573 | 28.6946 | 0.7698 | 0.2511 | 36.5526 |
| | 15 | 0.3921 | 28.4529 | 0.7611 | 0.2514 | 35.1040 |
| | 20 | 0.5305 | 28.3051 | 0.7557 | 0.2521 | 34.6182 |
| TTRD3-1Net-B | 5 | 0.1863 | 29.3176 | 0.7981 | 0.2578 | 40.4760 |
| | 10 | 0.3932 | 28.8538 | 0.7819 | 0.2487 | 34.0643 |
| | 15 | 0.6058 | 28.6528 | 0.7745 | 0.2472 | 32.3950 |
| | 20 | 0.8137 | 28.5171 | 0.7695 | 0.2465 | 31.8126 |
| TTRD3-2Net | 5 | 0.2529 | 29.3393 | 0.7999 | 0.2535 | 40.0347 |
| | 10 | 0.5365 | 28.8432 | 0.7808 | 0.2467 | 32.7252 |
| | 15 | 0.8125 | 28.6096 | 0.7739 | 0.2459 | 31.3301 |
| | 20 | 1.0911 | 28.4726 | 0.7686 | 0.2452 | 30.6984 |

### G. Estimation of Model Efficiency

To validate the inference efficiency of TTRD3, this study evaluates the model from two perspectives: parameter count (unit in M) and per-image inference time (unit in s). All tests were conducted under a unified NVIDIA RTX 4090 GPU environment to ensure fairness, with results visualized in Fig. 17. The $x$-axis represents parameter count and inference time, the $y$-axis denotes FID scores, and the color mapping of each method corresponds to LPIPS-based normalized scores (red indicates optimal performance).

Experimental results demonstrate that among DMs-based methods, the TTRD3-2Net achieves optimal performance metrics (FID: 32.7252) with only ten sampling steps. Its residual-noise dual prediction mechanism significantly reduces iterative requirements through complementary characteristics. Although TTRD3-2Net's parameter count (110.11M) is comparable to other DMs, the efficiency of dual diffusion enables superior performance with fewer steps, achieving a per-image inference time of 0.537 s. While slower than the pure residual prediction method ResShift (0.2713 s), TTRD3-2Net improves the FID metric by 45.3% compared with ResShift (FID: 59.8146). For a more lightweight option, TTRD3-1Net-A not only achieves faster inference (0.2573 s) than ResShift (0.2713 s) but also attains a 38.9% FID improvement. Notably, while current DM frameworks lag behind single-step inference methods (e.g., GAN-based SISR and RefSR) in computational efficiency due to iterative generation, they exhibit significant advantages in generative capability and visual quality metrics.

To further investigate the impact of sampling steps on TTRD3 performance, ablation studies were performed on

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:58:10 UTC from IEEE Xplore. Restrictions apply.

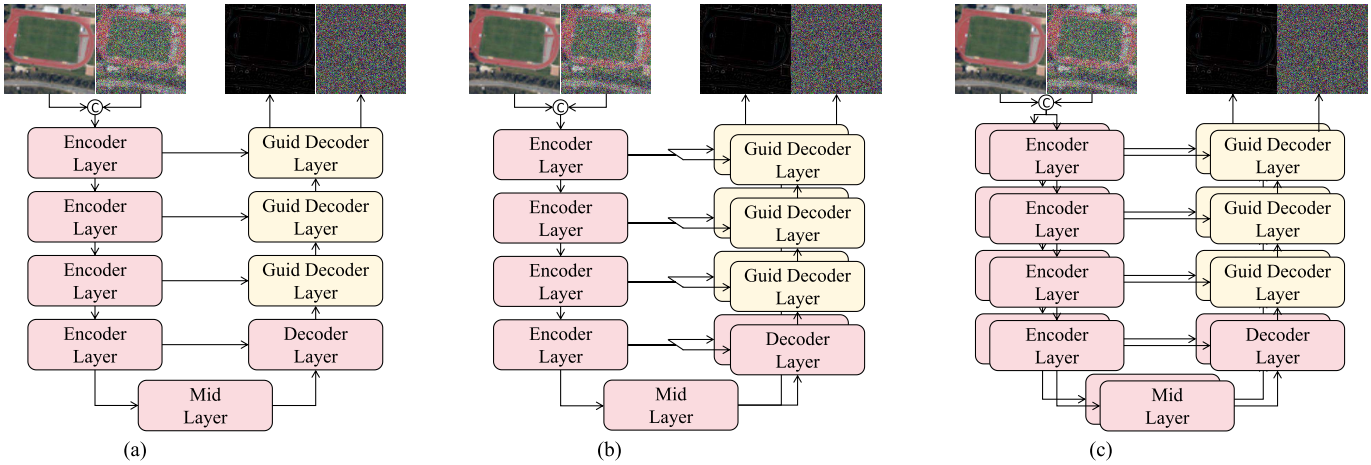



Fig. 16. Schematic of the three residual denoising prediction network structures. (a) TTRD3-1Net-A. (b) TTRD3-1Net-B. (c) TTRD3-2Net.

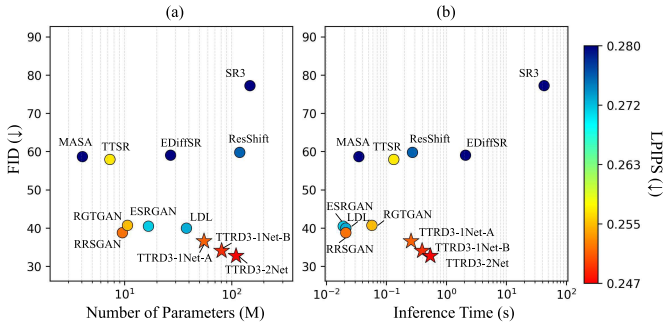



Fig. 17. Comparison chart of (a) model complexity and (b) inference efficiency, where the horizontal axis represents the number of parameters and single-image inference speed, the vertical axis represents the FID metric, and the color indicates the LPIPS metric.

TTRD3-1Net-A, TTRD3-1Net-B, and TTRD3-2Net using step counts of 5, 10, 15, and 20. Metrics and inference times were recorded on an NVIDIA RTX 4090 GPU (see Table VIII). Quantitative analysis showed that as sampling steps increased, SSIM and PSNR slightly declined, while perceptual quality metrics (LPIPS and FID) improved significantly, with inference times increasing approximately linearly. This finding underscores the necessity of progressive denoising for high-frequency detail recovery and provides critical guidance for parameter configuration: 5–10 steps balance quality and efficiency in time-sensitive scenarios, while 15–20 steps optimize visual fidelity for precision-critical tasks.

## VI. Conclusion

This study proposes TTRD3 for RSISR. The core RDDM framework integrates deterministic residual diffusion and stochastic noise diffusion to enhance texture diversity while preserving structural fidelity. To address the scarcity of high-frequency priors, MFAB fuses multiscale features extracted by convolutional kernels of different scales, while STTG adaptively transfers semantically aligned high-frequency textures from Ref images through a soft-hard attention collaborative mechanism. Evaluations demonstrate that TTRD3 outperforms SOTA methods in both fidelity and perceptual quality. Three architectural variants (TTRD3-1Net-A/B and TTRD3-2Net) are designed to flexibly adapt to different hardware conditions and task requirements, achieving a balance between high-precision reconstruction and fast processing.

Current limitations of TTRD3 include the fixed bicubic degradation assumption, which fails to effectively model composite degradations in real RS scenarios. Future work will extend TTRD3 to blind SR by incorporating degradation-aware mechanisms and kernel estimation.

## References

[1] M. Krichen, M. S. Abdalzaher, M. Elwekeil, and M. M. Fouda, "Managing natural disasters: An analysis of technological advancements, opportunities, and challenges," *Internet Things Cyber-Phys. Syst.*, vol. 4, pp. 99–109, Jan. 2024.

[2] H. Shirmard, E. Farahbakhsh, R. D. Müller, and R. Chandra, "A review of machine learning in processing remote sensing data for mineral exploration," *Remote Sens. Environ.*, vol. 268, Jan. 2022, Art. no. 112750.

[3] T. Bai et al., "Deep learning for change detection in remote sensing: A review," *Geo-Spatial Inf. Sci.*, vol. 26, no. 3, pp. 262–288, 2022.

[4] Q. Yuan et al., "Deep learning in environmental remote sensing: Achievements and challenges," *Remote Sens. Environ.*, vol. 241, May 2020, Art. no. 111716.

[5] X. Wang et al., "A review of image super-resolution approaches based on deep learning and applications in remote sensing," *Remote Sens.*, vol. 14, no. 21, p. 5423, Oct. 2022.

[6] P. Wang, B. Bayram, and E. Sertel, "A comprehensive review on deep learning based remote sensing image super-resolution methods," *Earth-Sci. Rev.*, vol. 232, Sep. 2022, Art. no. 104110.

[7] D. C. Lepcha, B. Goyal, A. Dogra, and V. Goyal, "Image super-resolution: A comprehensive review, recent trends, challenges and applications," *Inf. Fusion*, vol. 91, pp. 230–260, Mar. 2023.

[8] H. Huan et al., "End-to-end super-resolution for remote-sensing images using an improved multi-scale residual network," *Remote Sens.*, vol. 13, no. 4, p. 666, Feb. 2021.

[9] W. Ma, Z. Pan, J. Guo, and B. Lei, "Achieving super-resolution remote sensing images via the wavelet transform combined with the recursive res-net," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 6, pp. 3512–3527, Jun. 2019.

[10] S. Lei, Z. Shi, and W. Mo, "Transformer-based multistage enhancement for remote sensing image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5615611.

[11] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[12] T. Benzenati, Y. Kessentini, and A. Kallel, "Pansharpening approach via two-stream detail injection based on relativistic generative adversarial networks," *Expert Syst. Appl.*, vol. 188, Feb. 2022, Art. no. 115996.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:58:10 UTC from IEEE Xplore. Restrictions apply.

[13] S. Seo et al., "UPSNet: Unsupervised pan-sharpening network with registration learning between panchromatic and multi-spectral images," *IEEE Access*, vol. 8, pp. 201199–201217, 2020.

[14] Z. Tu, X. Yang, X. He, J. Yan, and T. Xu, "RGTGAN: Reference-based gradient-assisted texture-enhancement GAN for remote sensing super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5607221.

[15] R. Dong et al., "Building bridges across spatial and temporal resolutions: Reference-based super-resolution via change priors and conditional diffusion model," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 27684–27694.

[16] J.-Y. Zhu, T. Park, P. Isola, and A. A. Efros, "Unpaired image-to-image translation using cycle-consistent adversarial networks," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2242–2251.

[17] Y. Lecun, L. Bottou, Y. Bengio, and P. Haffner, "Gradient-based learning applied to document recognition," *Proc. IEEE*, vol. 86, no. 11, pp. 2278–2324, 1998.

[18] I. Goodfellow et al., "Generative adversarial networks," *Commun. ACM*, vol. 63, no. 11, pp. 139–144, 2020.

[19] F.-A. Croitoru, V. Hondru, R. T. Ionescu, and M. Shah, "Diffusion models in vision: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 9, pp. 10850–10869, Sep. 2023.

[20] X. Wang, L. Sun, A. Chehri, and Y. Song, "A review of GAN-based super-resolution reconstruction for optical remote sensing images," *Remote Sens.*, vol. 15, no. 20, p. 5062, Oct. 2023.

[21] C. Dong, C. C. Loy, K. He, and X. Tang, "Image super-resolution using deep convolutional networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 38, no. 2, pp. 295–307, Feb. 2016.

[22] J. Kim, J. K. Lee, and K. M. Lee, "Accurate image super-resolution using very deep convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 1646–1654.

[23] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[24] C. Ledig et al., "Photo-realistic single image super-resolution using a generative adversarial network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 105–114.

[25] T. Tong, G. Li, X. Liu, and Q. Gao, "Image super-resolution using dense skip connections," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 4799–4807.

[26] J. Kim, J. K. Lee, and K. M. Lee, "Deeply-recursive convolutional network for image super-resolution," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 1637–1645.

[27] Y. Tai, J. Yang, and X. Liu, "Image super-resolution via deep recursive residual network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 3147–3155.

[28] N. Ahn, B. Kang, and K.-A. Sohn, "Fast, accurate, and lightweight super-resolution with cascading residual network," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 252–268.

[29] Y. Zhang, K. Li, K. Li, L. Wang, B. Zhong, and Y. Fu, "Image super-resolution using very deep residual channel attention networks," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 286–301.

[30] T. Dai, J. Cai, Y. Zhang, S.-T. Xia, and L. Zhang, "Second-order attention network for single image super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 11065–11074.

[31] J. Liang, J. Cao, G. Sun, K. Zhang, L. Van Gool, and R. Timofte, "SwinIR: Image restoration using Swin transformer," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshops (ICCVW)*, Oct. 2021, pp. 1833–1844.

[32] F. Yang, H. Yang, J. Fu, H. Lu, and B. Guo, "Learning texture transformer network for image super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 5791–5800.

[33] Z. Lu, J. Li, H. Liu, C. Huang, L. Zhang, and T. Zeng, "Transformer for single image super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2022, pp. 457–466.

[34] X. Zhang, H. Zeng, S. Guo, and L. Zhang, "Efficient long-range attention network for image super-resolution," in *Proc. Eur. Conf. Comput. Vis.* Tel Aviv, Israel: Springer, Oct. 2022, pp. 649–667.

[35] C. Ren, X. He, L. Qing, Y. Wu, and Y. Pu, "Remote sensing image recovery via enhanced residual learning and dual-luminance scheme," *Knowl.-Based Syst.*, vol. 222, Jun. 2021, Art. no. 107013.

[36] J. Wang, Y. Wu, L. Wang, L. Wang, O. Alfarraj, and A. Tolba, "Lightweight feedback convolution neural network for remote sensing images super-resolution," *IEEE Access*, vol. 9, pp. 15992–16003, 2021.

[37] X. Kang, P. Duan, J. Li, and S. Li, "Efficient Swin transformer for remote sensing image super-resolution," *IEEE Trans. Image Process.*, vol. 33, pp. 6367–6379, 2024.

[38] B. Chen, K. Chen, M. Yang, Z. Zou, and Z. Shi, "Heterogeneous mixture of experts for remote sensing image super-resolution," *IEEE Geosci. Remote Sens. Lett.*, vol. 22, 2025, Art. no. 6006405.

[39] J. Li, Q. Lv, W. Zhang, B. Zhu, G. Zhang, and Z. Tan, "Multi-attention multi-image super-resolution transformer (MAST) for remote sensing," *Remote Sens.*, vol. 15, no. 17, p. 4183, Aug. 2023.

[40] Y. Xiao, Q. Yuan, K. Jiang, J. He, C.-W. Lin, and L. Zhang, "TTST: A top-$k$ token selective transformer for remote sensing image super-resolution," *IEEE Trans. Image Process.*, vol. 33, pp. 738–752, 2024.

[41] M. S. M. Sajjadi, B. Scholkopf, and M. Hirsch, "EnhanceNet: Single image super-resolution through automated texture synthesis," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 4491–4500.

[42] X. Wang et al., "ESRGAN: Enhanced super-resolution generative adversarial networks," in *Proc. Eur. Conf. Comput. Vis. (ECCV) workshops*, 2019, pp. 63–79.

[43] X. Wang, L. Xie, C. Dong, and Y. Shan, "Real-ESRGAN: Training real-world blind super-resolution with pure synthetic data," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshops (ICCVW)*, Oct. 2021, pp. 1905–1914.

[44] H. Zhang, I. Goodfellow, D. Metaxas, and A. Odena, "Self-attention generative adversarial networks," in *Proc. Int. Conf. Mach. Learn.*, 2019, pp. 7354–7363.

[45] J. Liang, H. Zeng, and L. Zhang, "Details or artifacts: A locally discriminative learning approach to realistic image super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 5657–5666.

[46] M. S. Moustafa and S. A. Sayed, "Satellite imagery super-resolution using squeeze-and-excitation-based GAN," *Int. J. Aeronaut. Space Sci.*, vol. 22, no. 6, pp. 1481–1492, Dec. 2021.

[47] S. Jia, Z. Wang, Q. Li, X. Jia, and M. Xu, "Multiattention generative adversarial network for remote sensing image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5624715.

[48] Y. Li et al., "Single-image super-resolution for remote sensing images using a deep generative adversarial network with local and global attention mechanisms," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 3000224.

[49] Y. Xu, W. Luo, A. Hu, Z. Xie, X. Xie, and L. Tao, "TE-SAGAN: An improved generative adversarial network for remote sensing super-resolution images," *Remote Sens.*, vol. 14, no. 10, p. 2425, May 2022.

[50] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," in *Proc. NIPS*, vol. 33, 2020, pp. 6840–6851.

[51] J. Song, C. Meng, and S. Ermon, "Denoising diffusion implicit models," 2020, *arXiv:2010.02502*.

[52] J. Liu, Q. Wang, H. Fan, Y. Wang, Y. Tang, and L. Qu, "Residual denoising diffusion models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 2773–2783.

[53] P. Dhariwal and A. Nichol, "Diffusion models beat GANs on image synthesis," in *Proc. Adv. Neural Inf. Process. Syst.*, 2021, pp. 8780–8794.

[54] J. Ho and T. Salimans, "Classifier-free diffusion guidance," 2022, *arXiv:2207.12598*.

[55] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, "High-resolution image synthesis with latent diffusion models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun.~2022, pp. 10684–10695.

[56] C. Saharia, J. Ho, W. Chan, T. Salimans, D. J. Fleet, and M. Norouzi, "Image super-resolution via iterative refinement," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 4, pp. 4713–4726, Apr. 2022.

[57] H. Li et al., "SRDiff: Single image super-resolution with diffusion probabilistic models," *Neurocomputing*, vol. 479, pp. 47–59, Mar. 2022.

[58] Z. Yue, J. Wang, and C. C. Loy, "ResShift: Efficient diffusion model for image super-resolution by residual shifting," in *Proc. Adv. Neural Inf. Process. Syst.*, 2023, pp. 13294–13307.

[59] A. M. Ali, B. Benjdira, A. Koubaa, W. Boulila, and W. El-Shafai, "TESR: Two-stage approach for enhancement and super-resolution of remote sensing images," *Remote Sens.*, vol. 15, no. 9, p. 2346, Apr. 2023.

[60] Y. Xiao, Q. Yuan, K. Jiang, J. He, X. Jin, and L. Zhang, "EDiffSR: An efficient diffusion probabilistic model for remote sensing image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5601514.

[61] F. Meng et al., "A conditional diffusion model with fast sampling strategy for remote sensing image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5408616.

[62] J. Wang, Z. Yue, S. Zhou, K. C. K. Chan, and C. C. Loy, "Exploiting diffusion prior for real-world image super-resolution," *Int. J. Comput. Vis.*, vol. 132, no. 12, pp. 5929–5949, Dec. 2024.


Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:58:10 UTC from IEEE Xplore. Restrictions apply.

[63] H. Wu, J. Mo, X. Sun, and J. Ma, "Latent diffusion, implicit amplification: Efficient continuous-scale super-resolution for remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5625417.
[64] S. Donike, C. Aybar, L. Gómez-Chova, and F. Kalaitzis, "Trustworthy super-resolution of multispectral Sentinel-2 imagery with latent diffusion," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 6940–6952, 2025.
[65] C. Zhu, Y. Liu, S. Huang, and F. Wang, "Taming a diffusion model to revitalize remote sensing image super-resolution," *Remote Sens.*, vol. 17, no. 8, p. 1348, Apr. 2025.
[66] Z. Zhang, Z. Wang, Z. Lin, and H. Qi, "Image super-resolution by neural texture transfer," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 7982–7991.
[67] L. Lu, W. Li, X. Tao, J. Lu, and J. Jia, "MASA-SR: Matching acceleration and spatial adaptation for reference-based image super-resolution," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 6368–6377.
[68] H. Sun et al., "CoSeR: Bridging image and language for cognitive super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 25868–25878.
[69] R. Dong, L. Zhang, and H. Fu, "RRSGAN: Reference-based super-resolution for remote sensing image," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5601117.
[70] Y. Meng, W. Li, S. Lei, Z. Zou, and Z. Shi, "Large-factor super-resolution of remote sensing images with spectra-guided generative adversarial networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5634111.
[71] C. Wang and W. Sun, "Semantic guided large scale factor remote sensing image super-resolution with generative diffusion prior," *ISPRS J. Photogramm. Remote Sens.*, vol. 220, pp. 125–138, Feb. 2025.
[72] Y. Bengio, N. Léonard, and A. Courville, "Estimating or propagating gradients through stochastic neurons for conditional computation," 2013, *arXiv:1308.3432*.
[73] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, no. 4, pp. 600–612, Apr. 2004.
[74] R. Zhang, P. Isola, A. A. Efros, E. Shechtman, and O. Wang, "The unreasonable effectiveness of deep features as a perceptual metric," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 586–595.
[75] N. B. Bynagari, "GANs trained by a two time-scale update rule converge to a local Nash equilibrium," *Asian J. Appl. Sci. Eng.*, vol. 8, no. 1, pp. 25–34, Apr. 2019.
[76] A. Mittal, R. Soundararajan, and A. C. Bovik, "Making a 'completely blind' image quality analyzer," *IEEE Signal Process. Lett.*, vol. 20, no. 3, pp. 209–212, Mar. 2012.



**Yide Liu** received the Bachelor of Engineering degree in automation from Yanshan University, Qinhuangdao, China, in 2022. He is currently pursuing the joint Ph.D. degree in signal and information processing with Changchun Institute of Optics, Fine Mechanics and Physics, Chinese Academy of Sciences, Changchun, China, and the University of Chinese Academy of Sciences, Beijing, China.

His research interests focus on deep learning and its applications in remote sensing image processing, particularly in image super-resolution and intelligent interpretation.



**Haijiang Sun** received the Ph.D. degree in electronic engineering from Changchun Institute of Optics, Fine Mechanics and Physics (CIOMP), Chinese Academy of Sciences, Changchun, China, in 2010.

He is currently a Researcher with the Perception and Display Laboratory, CIOMP, Chinese Academy of Sciences. His research interests include target recognition and tracking, high-definition video enhancement, and display.

Dr. Sun is an Associate Editor of the *Liquid Crystal and Reality and Optical Precision Engineering*.



**Xiaowen Zhang** is currently pursuing the Ph.D. degree with Changchun Institute of Optics, Fine Mechanics and Physics, Chinese Academy of Sciences, Changchun, China.

His research interests include computer vision, remote sensing image object detection, and object tracking.



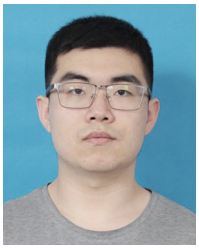
**Qiaoyuan Liu** received the master's and Ph.D. degrees from the Department of Computer Science and Technology, Northeast Normal University, Changchun, China, in 2016 and 2019, respectively.

She is currently an Associate Researcher with Changchun Institute of Optics, Fine Mechanics and Physics (CIOMP), Chinese Academy of Sciences, Changchun. Her research interests including intelligent remote sensing image processing, computer vision, and pattern recognition.



**Zhouchang Chen** received the B.Eng. degree in automation from Fuzhou University, Fuzhou, China, in 2022. He is currently pursuing the joint Ph.D. degree in signal and information processing with the University of Chinese Academy of Sciences Beijing, China, and Changchun Institute of Optical Precision Machinery and Physics, Chinese Academy of Sciences, Changchun, China.

His research interests include deep learning, intelligent interpretation of remote sensing images, and change detection.



**Chongzhuo Xiao** received the Bachelor of Science degree in robotics engineering from Beijing Information Science and Technology University, Beijing, China, in 2022. He is currently pursuing the joint Ph.D. degree in signal and information processing with Changchun Institute of Optics, Fine Mechanics and Physics, Chinese Academy of Sciences, Changchun, and the University of Chinese Academy of Sciences, Beijing.

His research interests focus on deep learning, image processing, and Pansharpening.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:58:10 UTC from IEEE Xplore. Restrictions apply.